INTRODUCTION TO COMPUTER HARDWARE

COMPUTER HARDWARE:

- The Physical Components of Computer are called Hardware.

List of Few Computer Hardware's:

1) Power Supply
2) Mother Board
3) Processor (CPU)
4) Heat Sink
5) RAM
6) Hard disk Drive
7) BIOS
8) Video Card
9) Sound Card
10) Network Card

Power Supply:

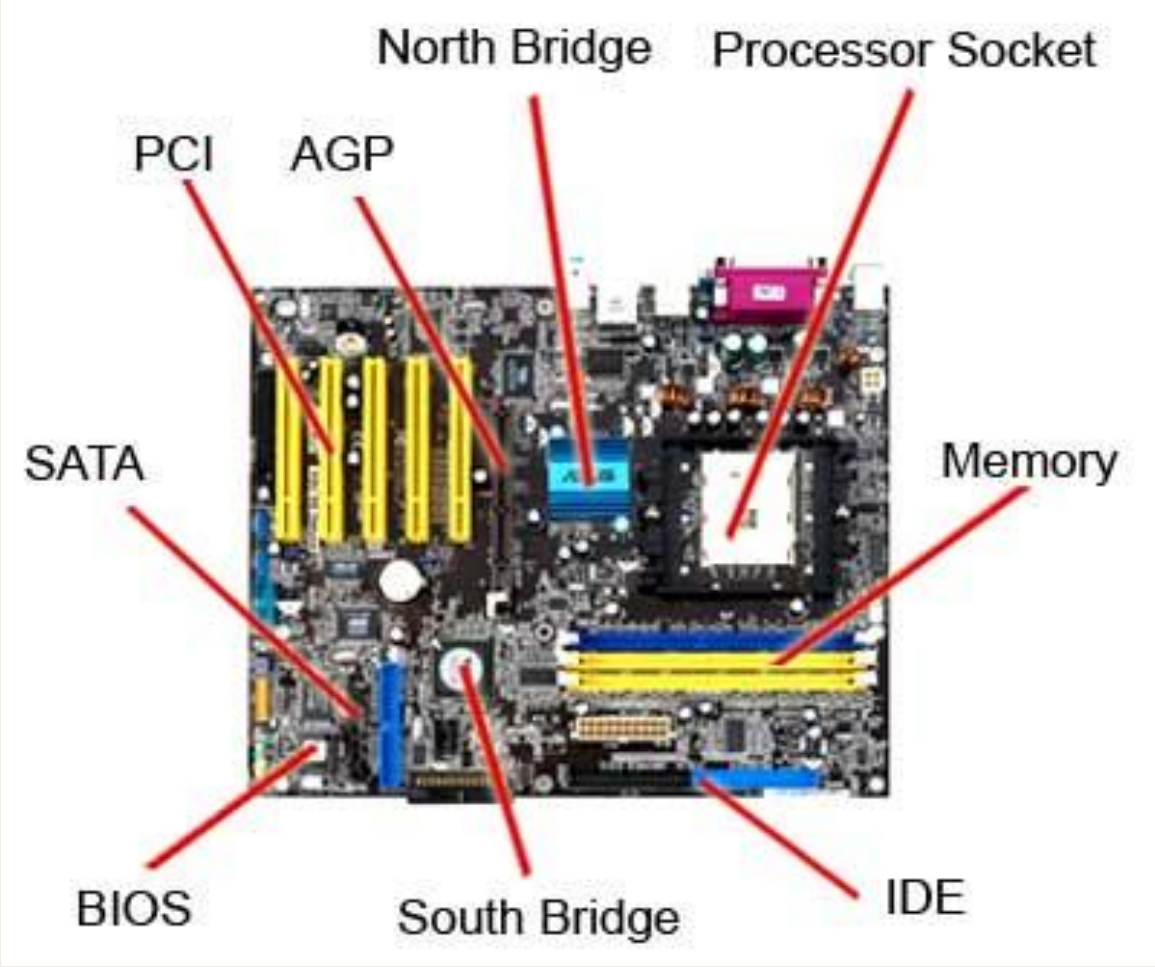
The power supply converts the Alternating current (AC) from your mains (110V input or 220V input) to the Direct current (DC) needed by the computer. In a PC, the power supply is the metal box usually found in a corner of the case.

Component	Requirement
AGP Video Card	30W – 50W
PCI Express Card	100W – 250W
Average PCI Card	5W – 10W
DVD/CD	20W – 30W
Hard Drive	15W – 30W
Case/CPU Fans	3W
Motherboard (without CPU or RAM)	50W – 150W
RAM	15W per 1GB
Processor	80W – 140W



Motherboard:

The motherboard is the key circuit board holding the essential processing parts of a computer. It allows all the parts of your computer to receive power and communicate with one another.



Processor (CPU):

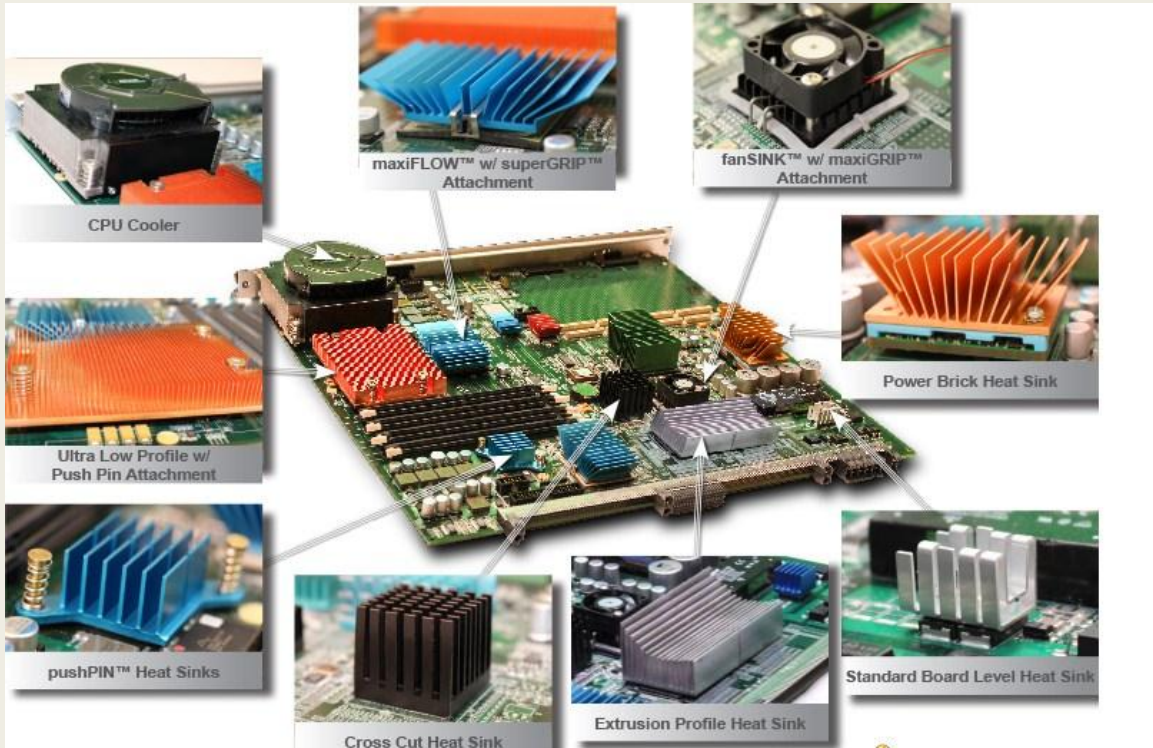
Stands for "Central Processing Unit." The CPU is the primary component of a computer that processes instructions. It processes the Data and produces output, which may stored by an application or displayed on the screen.





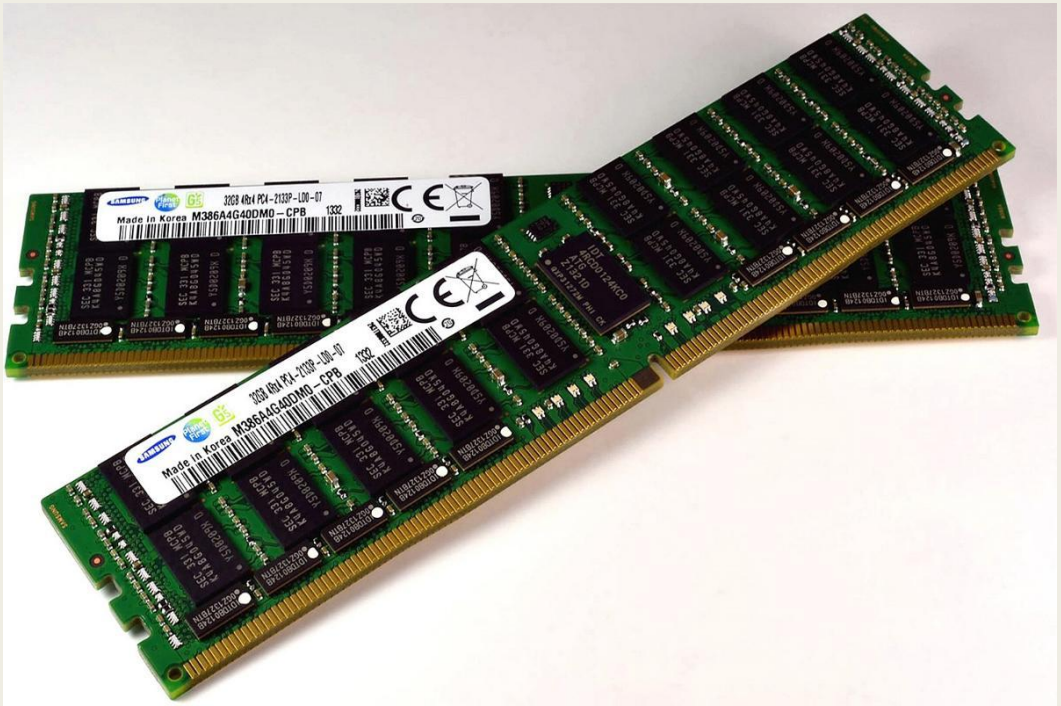
Heat Sinks/Fans:

As processors, graphics cards, RAM and other components in computers have increased in speed and power consumption, the amount of heat produced by these components as a side-effect of normal operation has also increased. These components need to be kept within a specified temperature range to prevent overheating, instability, malfunction and damage leading to a shortened component lifespan.



RAM (Random access memory):

RAM is a series of small cards or modules plugged into slots on the motherboard. The CPU can request any data in RAM. It is then **located, opened,** and **delivered** to the CPU for processing in a few **billionths of a second.** Since all the contents of RAM are erased when you turn off the computer, RAM is the **temporary** or **volatile** storage location for the computer.



Hard Disk Drive:

A hard disk drive (HDD) is a non-volatile computer storage device containing magnetic disks or platters rotating at high speeds. It is a secondary storage device used to store data permanently.





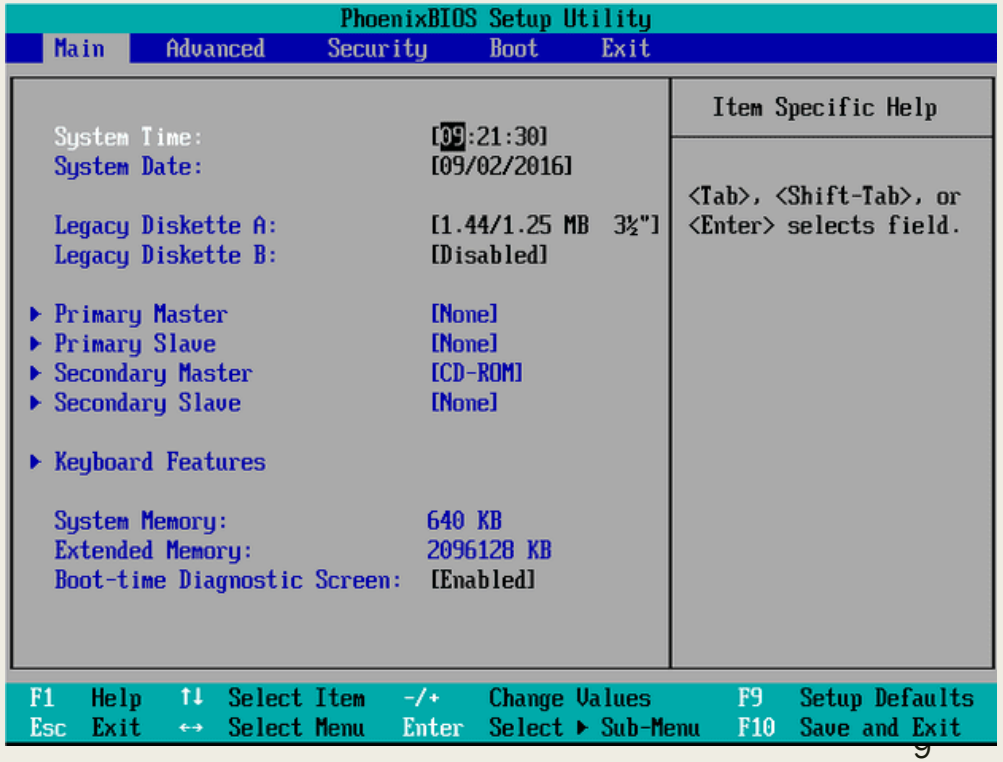
BIOS:

The basic input-output system (BIOS) is the first thing you see when you turn on your computer. The BIOS is special software that interfaces the major hardware components of your computer with the operating system. It is usually stored on a Flash memory chip on the motherboard (which has its own battery), but sometimes the chip is another type of ROM.

ROM

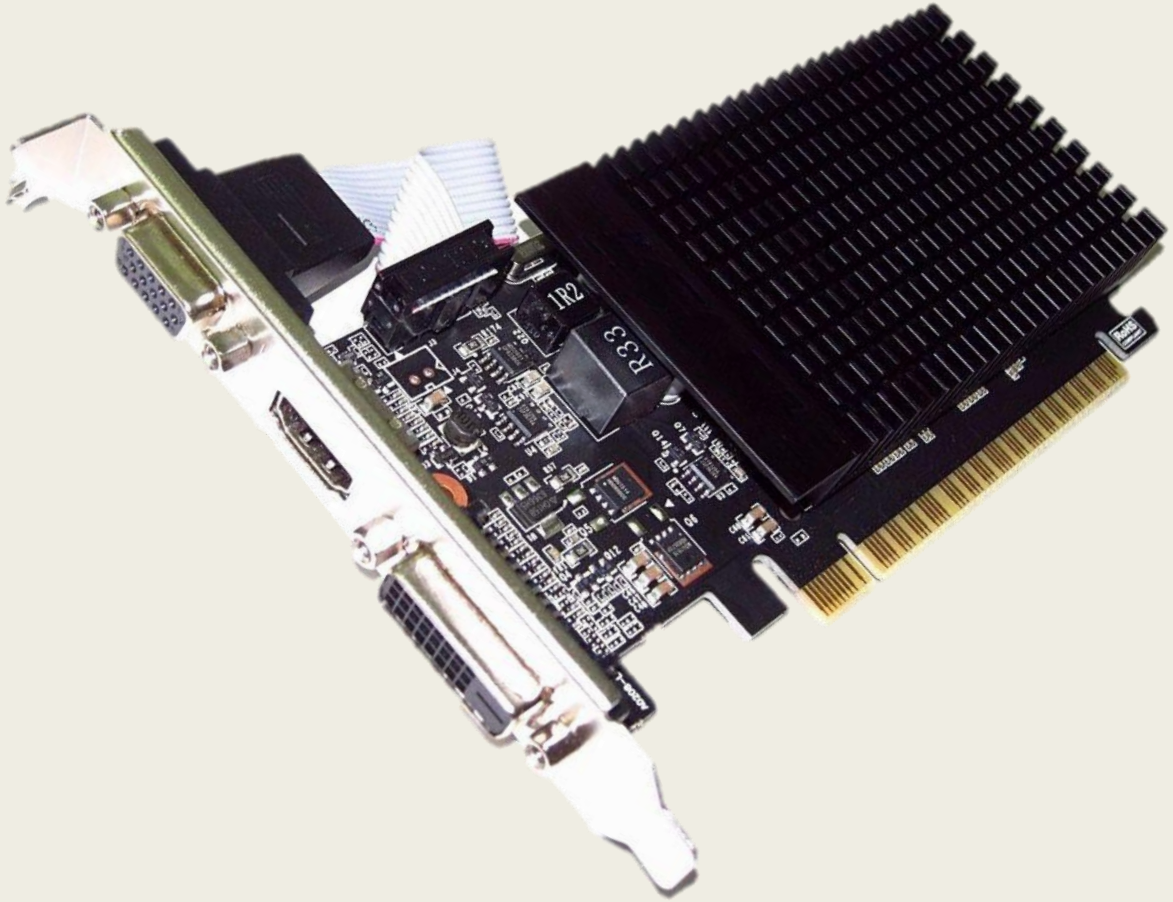


BIOS



Video (Graphics) Card:

A dedicated video card (or video adapter) is an expansion card installed inside your system unit to translate binary data received from the CPU or GPU into the images you view on your monitor. It is an alternative to the integrated graphics chip.



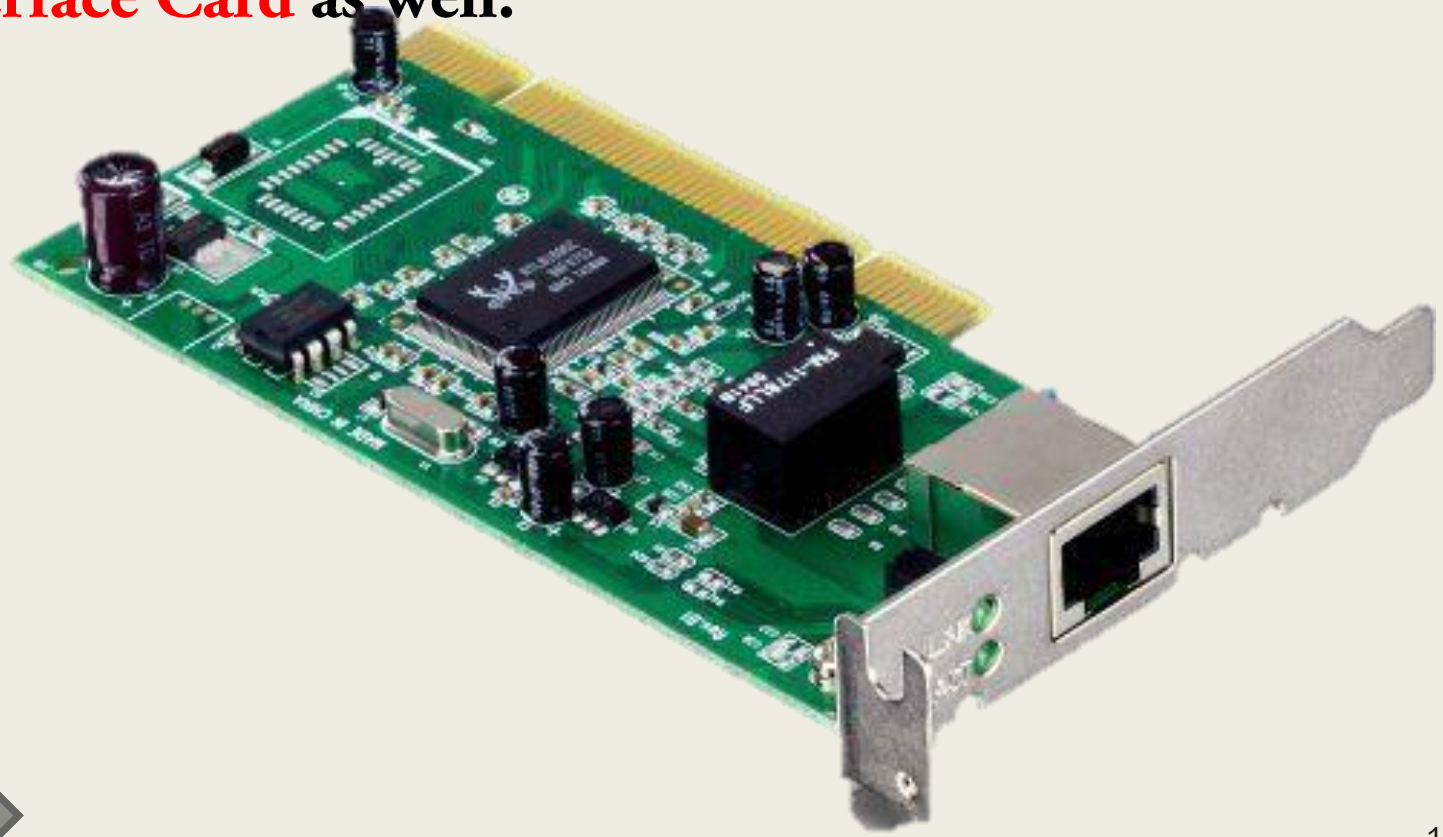
Sound Card:

Sound cards attached to the motherboard and enabled your computer to record and reproduce sounds. The ports on the sound card allow you to connect additional audio devices such as amplified speakers, headphones, microphones etc.



Network Card:

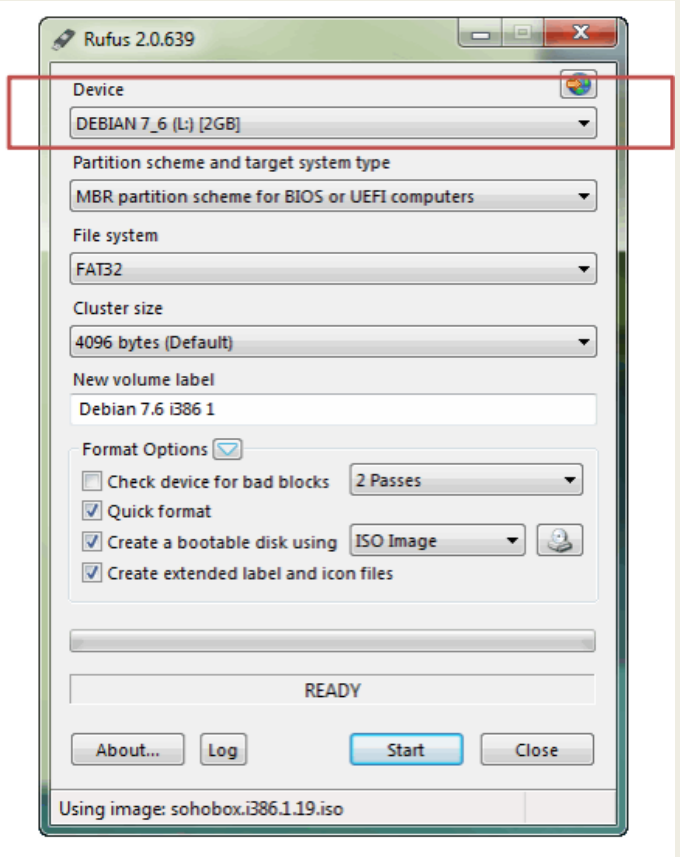
An Ethernet network requires that you install or attach network adapters to each computer or peripheral you want to connect to the network. Most computers come with Ethernet adapters preinstalled as network interface cards (NICs). If your computer does not have NIC so you can installed external Network Interface Card as well.



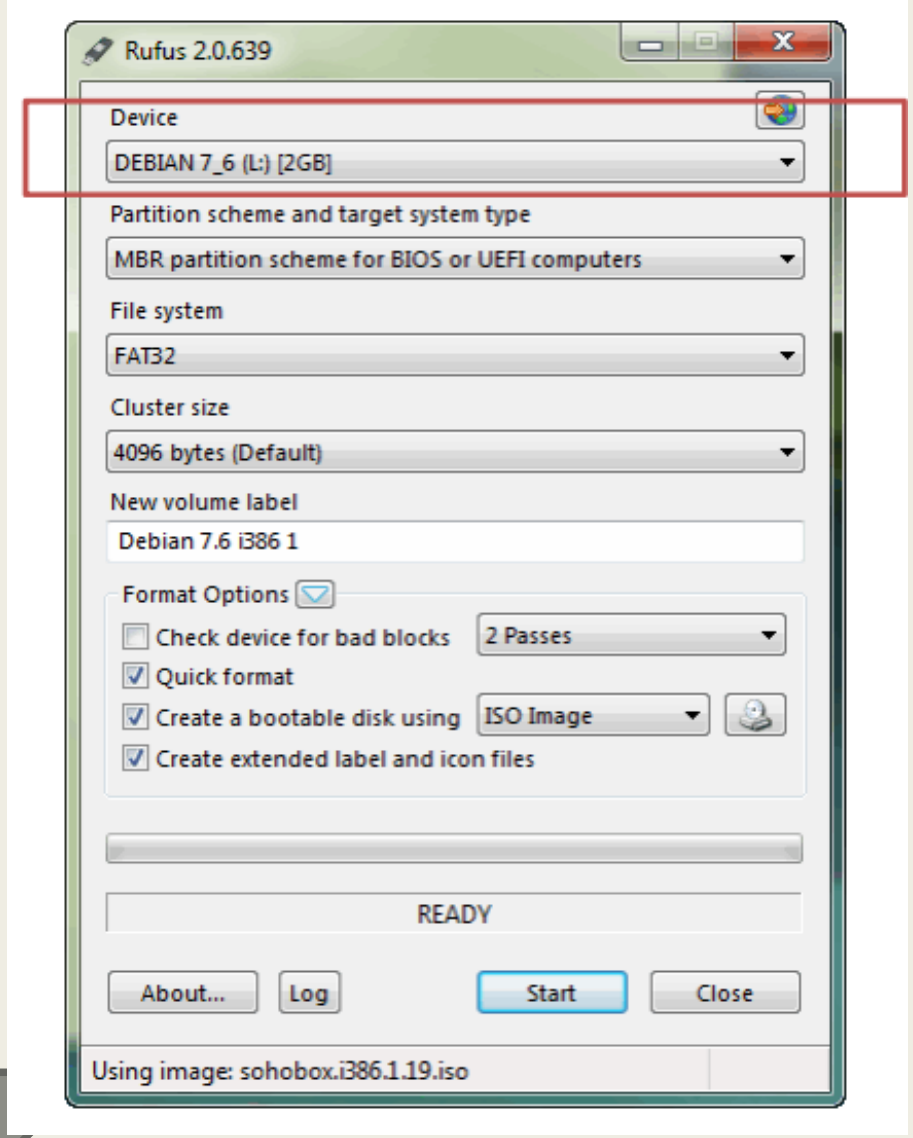
How to Make USB Drive Bootable Step by Step Procedure.

To create a bootable USB drive from a downloaded ISO file, we use a free software called RUFUS and we need an empty USB flash drive with minimum size of 4Gb Go in order to properly transfer your ISO image file on it.

- Run RUFUS on your Windows PC
- Choose in “device” your drive that contains your emp
  USB flash drive

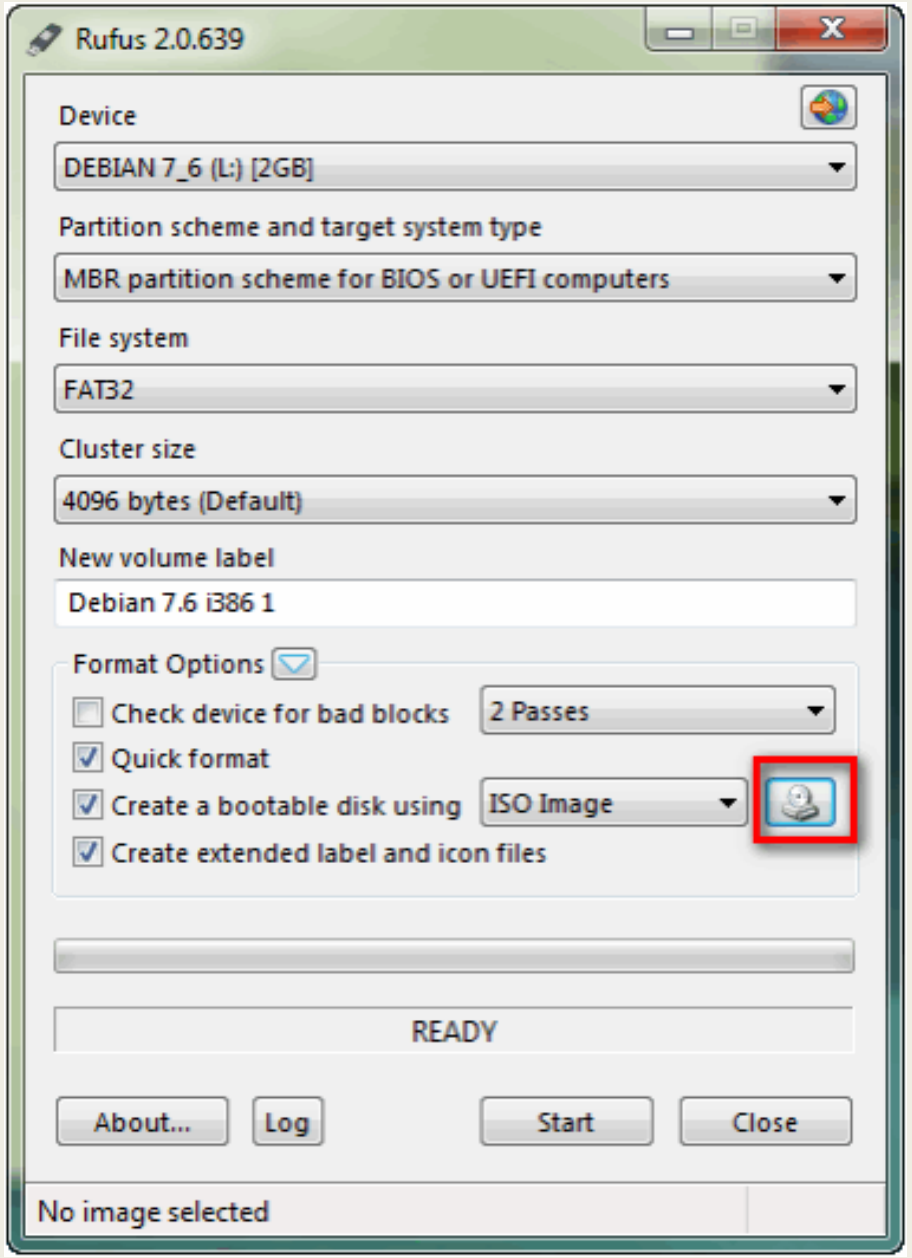


You can leave other settings by default or as shown in the screenshot. Make sure that the checkbox "create a bootable disk" is selected.



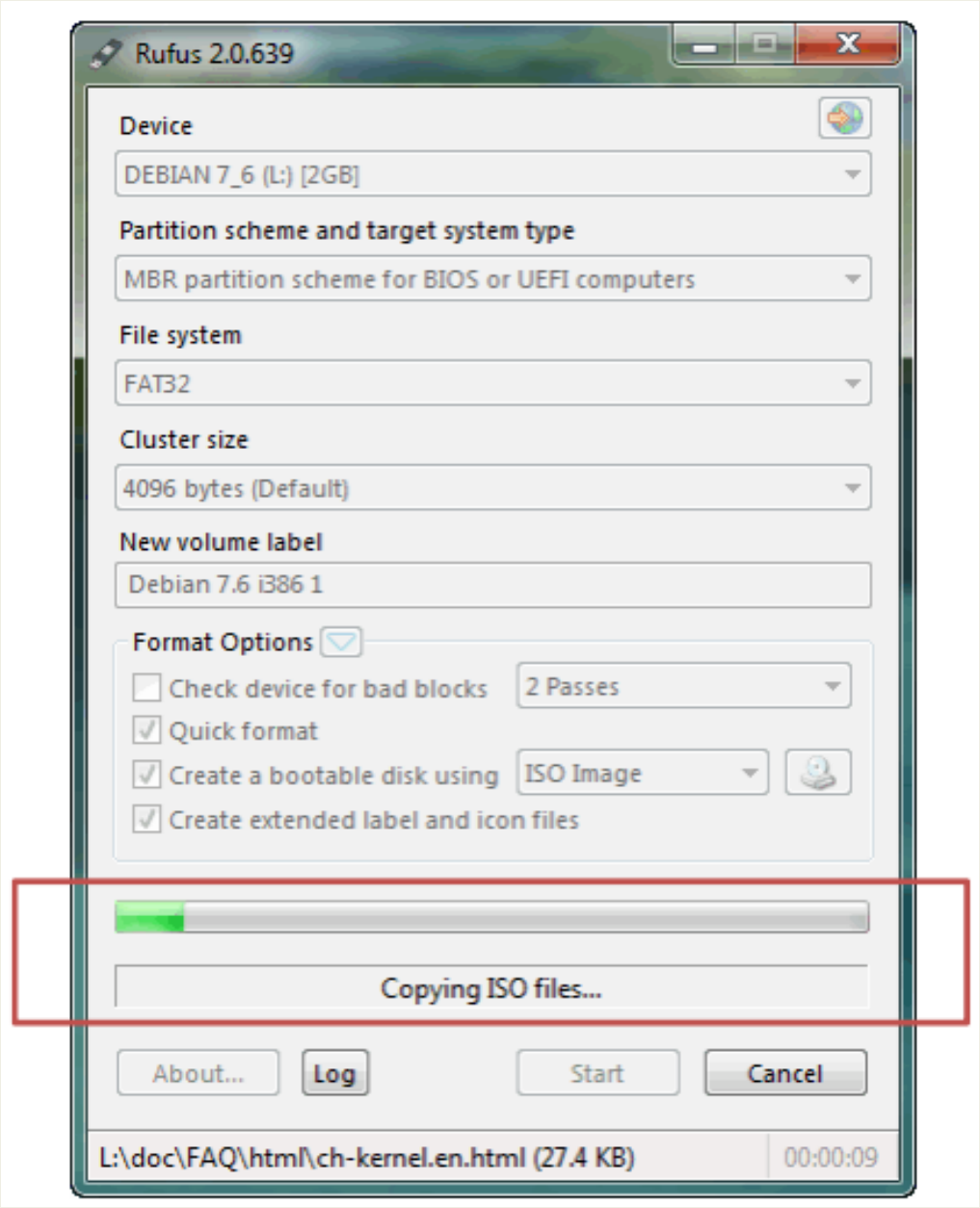
Select your ISO image file to be transferred on your USB flash drive, as shown below in the red rectangle.

Make sure to leave all other options as shown in the screenshot.

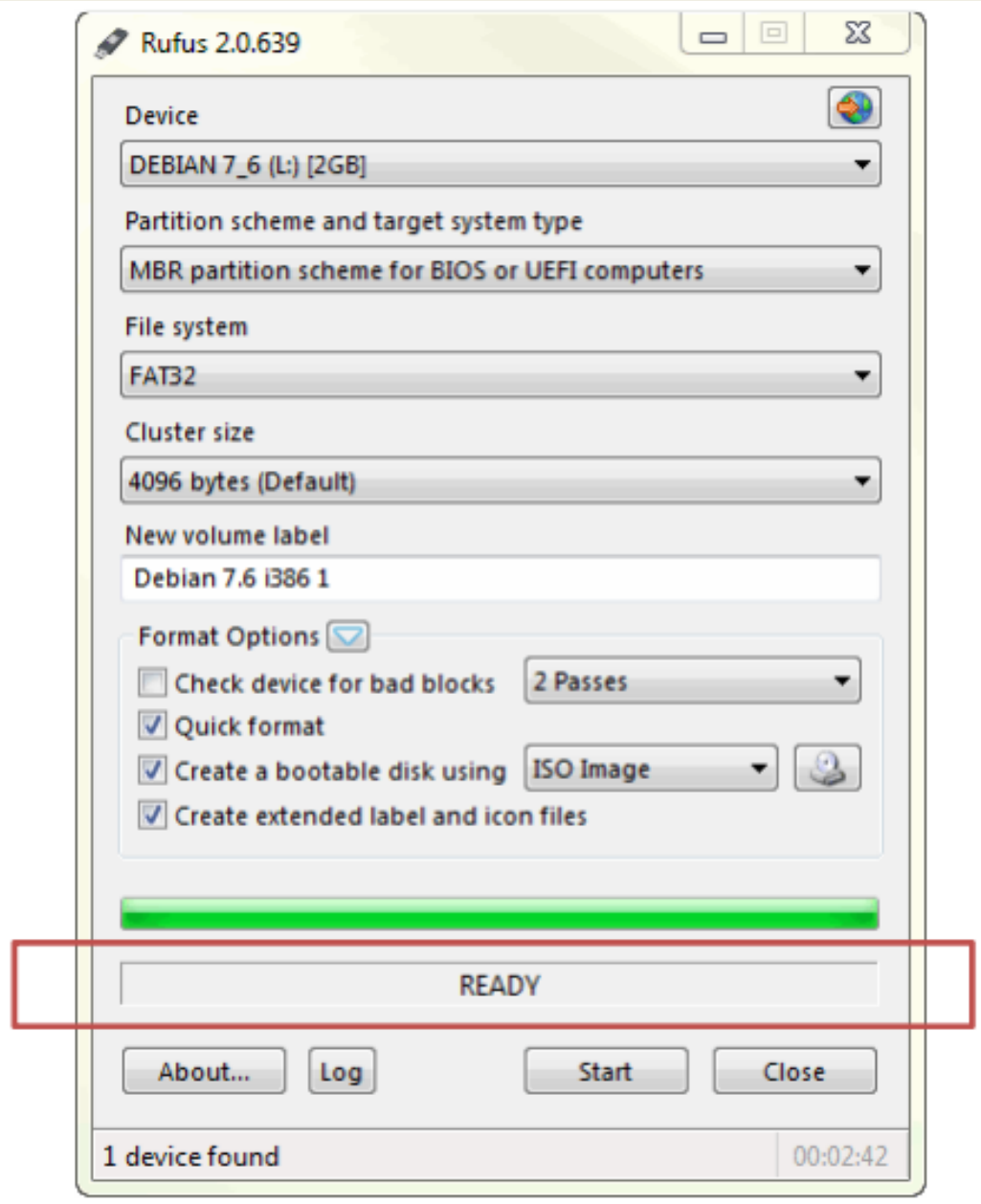


Then that's it, you can press the "Start" button to begin the transfer.

Your ISO will be transferred to your USB drive

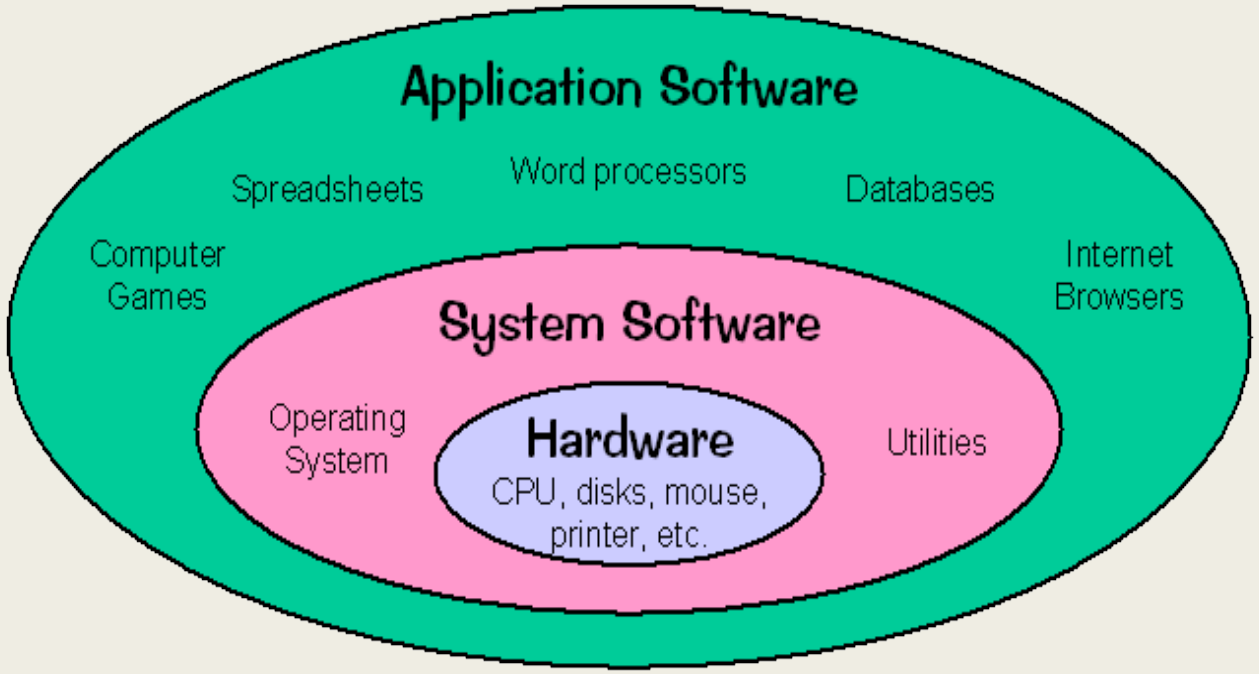


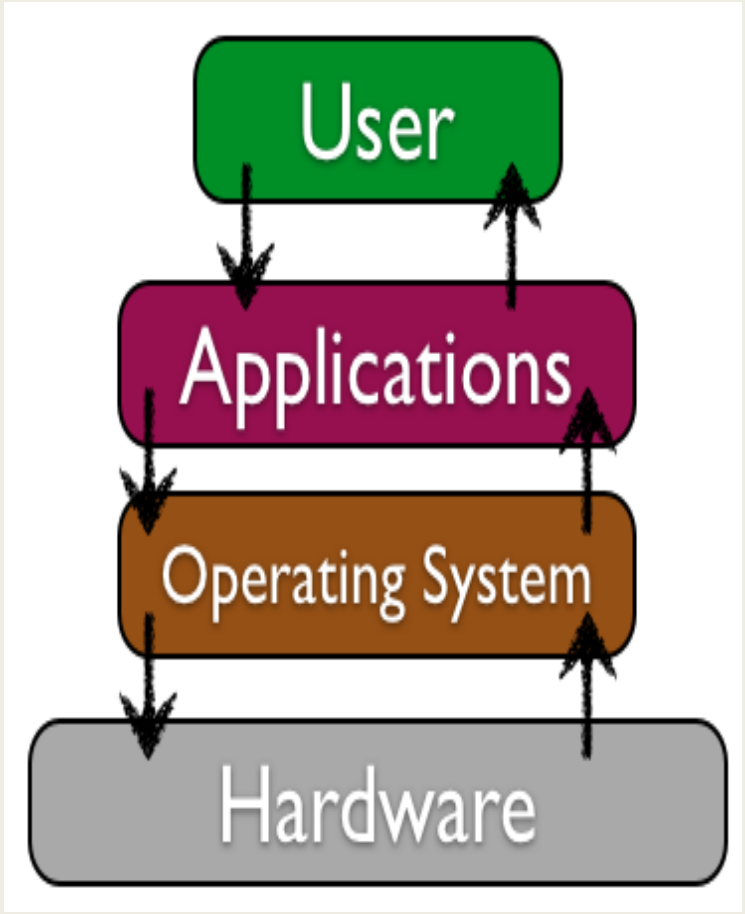
At the end of the installation, your USB flash drive is ready to use.



How to Install Windows 7 Operating System Step by Step Procedure.

Introduction Operating System





Operating System:

- The Operating System provides for the user to utilize the functions of a computer by managing the hardware and software in it. The image Above depicts how the system software and application software interact with the hardware.

The software which facilitate the interaction between human user and hardware is the Operating System. The Operating System provides instructions for installation and management of various application software. Not only that the Operating System manages all the input, output and computer memory too, which means that Operating System is the sole software which manages the whole.

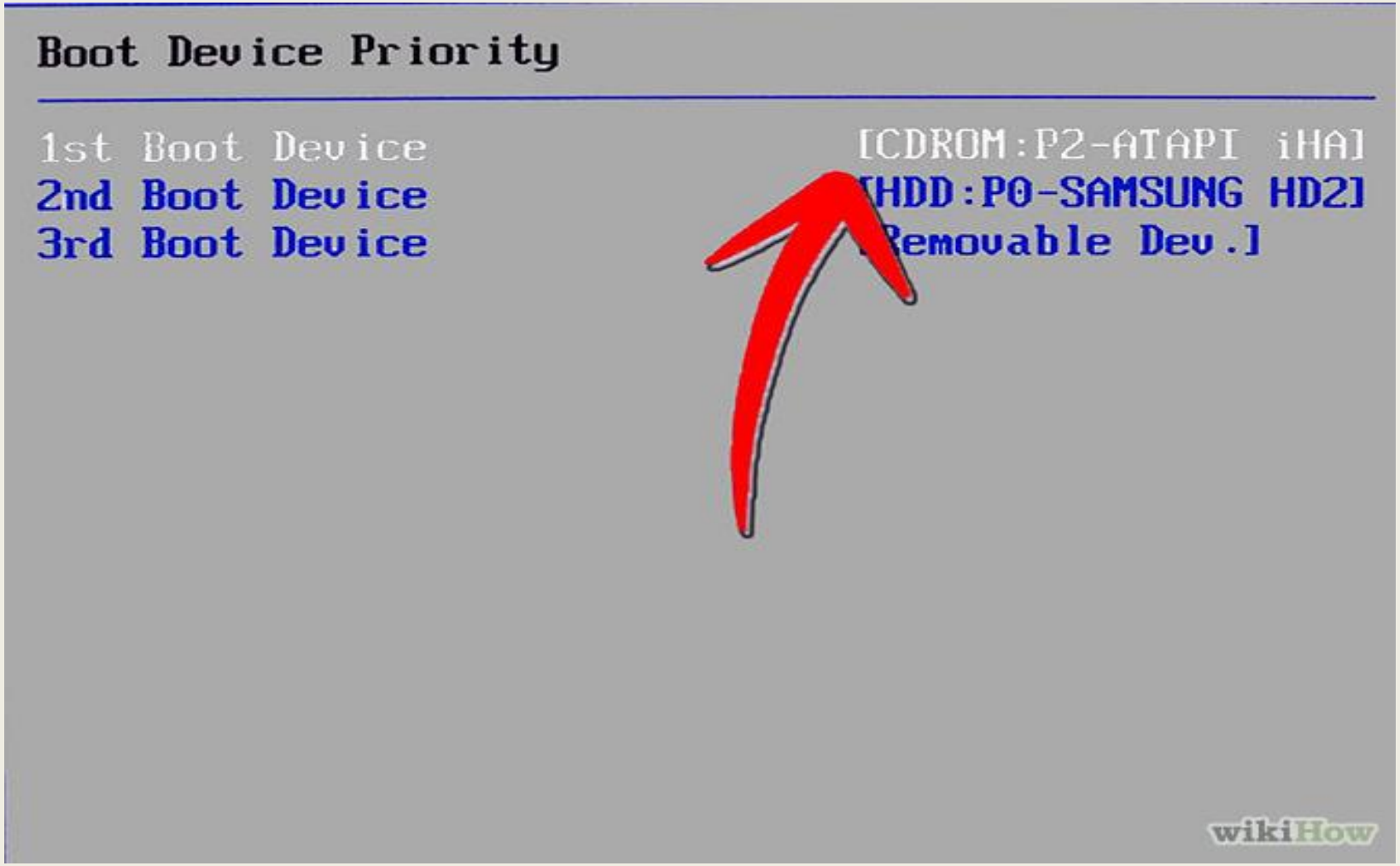
Minimum System Requirements for Windows 7

Part	Minimum Requirement
Processor (CPU)	1 GHz or faster CPU
Memory (RAM)	1 GB RAM (32-bit) or 2 GB RAM (64-bit)
Hard drive free space	16 GB available disk space (32-bit) or 20 GB (64-bit)
Disc drive	DVD drive
Graphics memory	128 MB or more

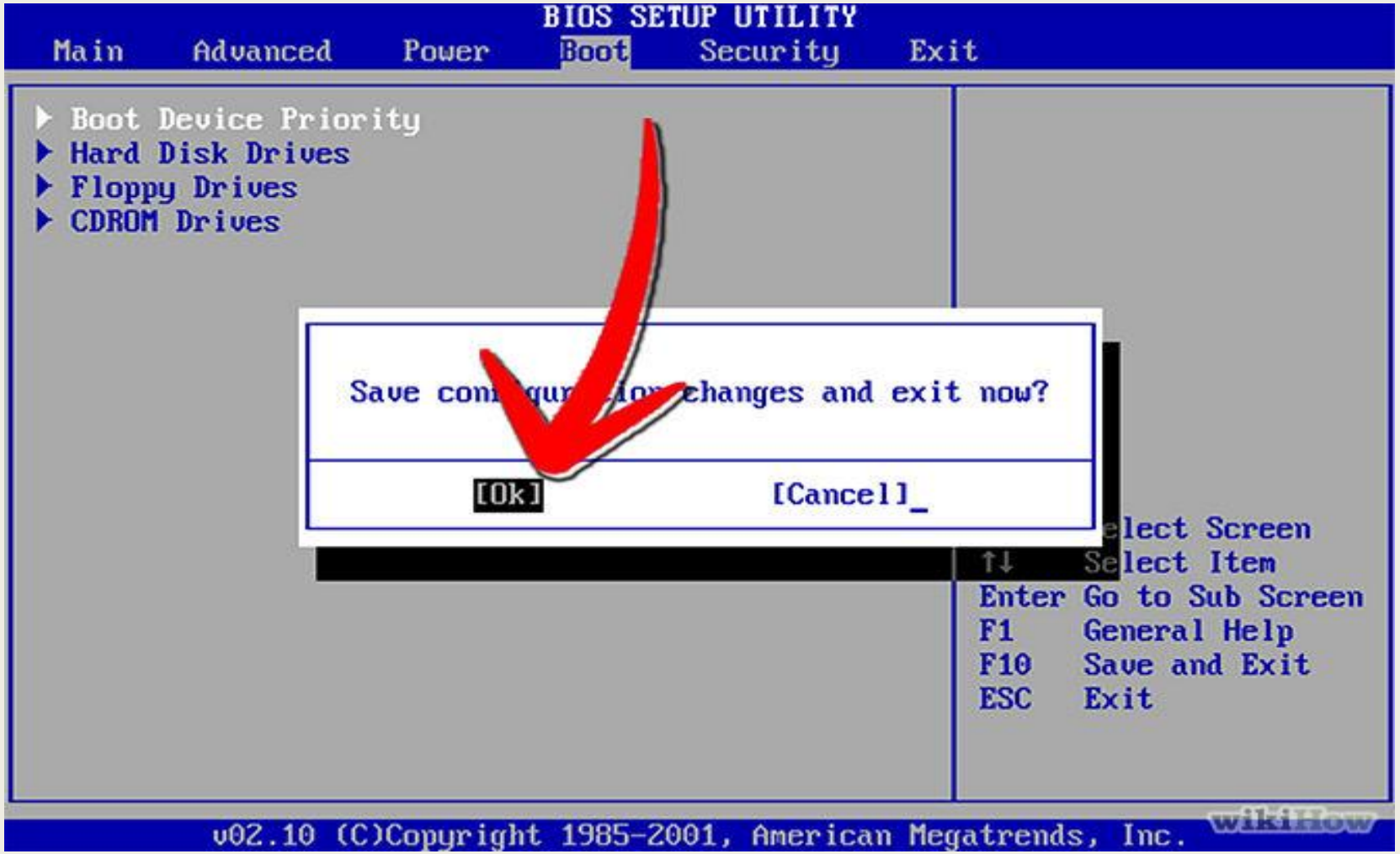
Operating system boot by:

1. Flash Drive (Bootable Pen Drive)
2. Bootable CD/DVD
3. Hard Disk

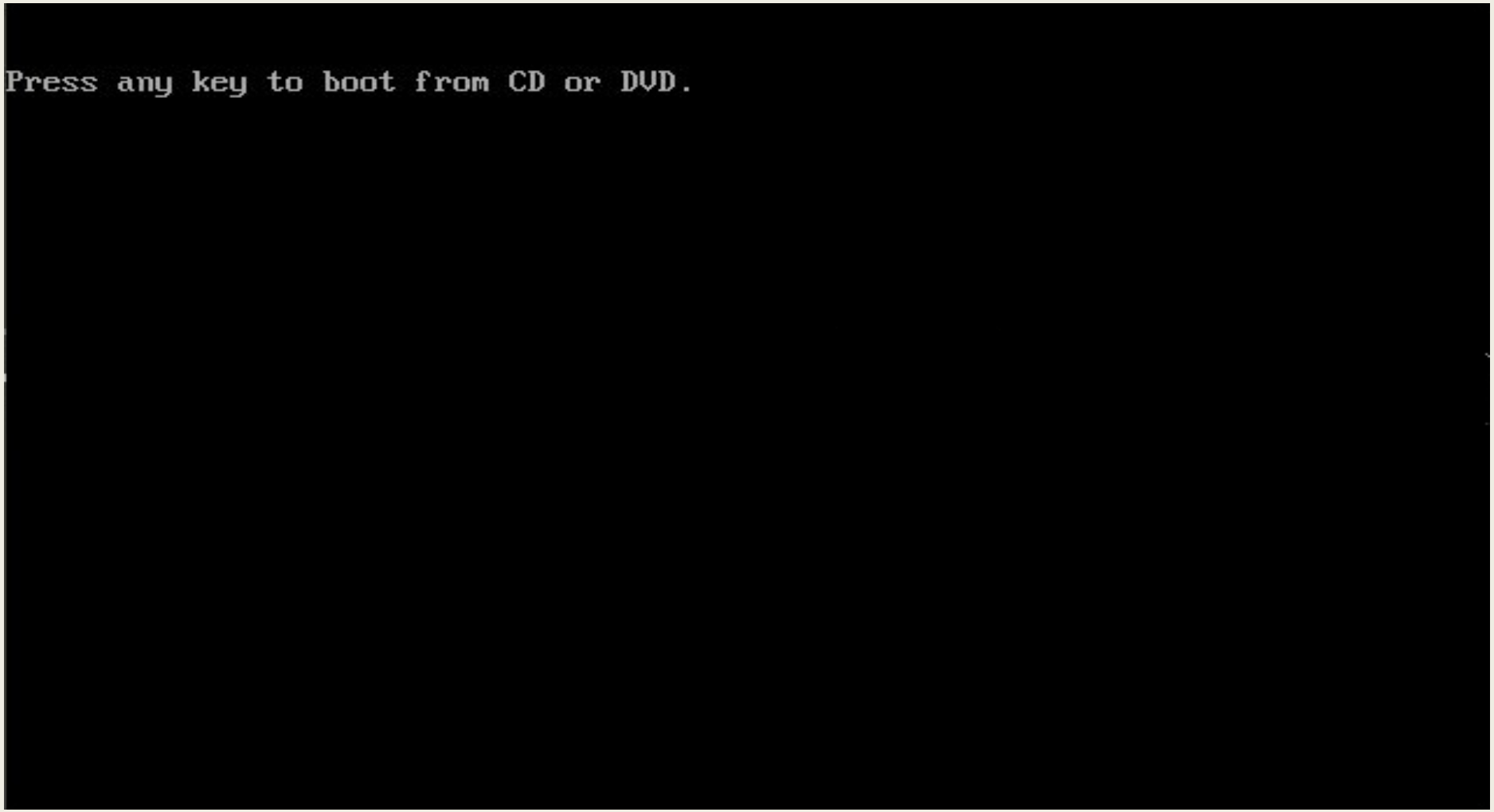
Turn your computer on and then press F2 on Dell's System (depend on your computer's mainboard) to enter the system BIOS



Go to Boot menu and Choose Boot From CD/DVD.



Load your Windows 7 DVD and boot it. It will now load the setup files. Press any key before five seconds.



Loading process



Booting process is in progress

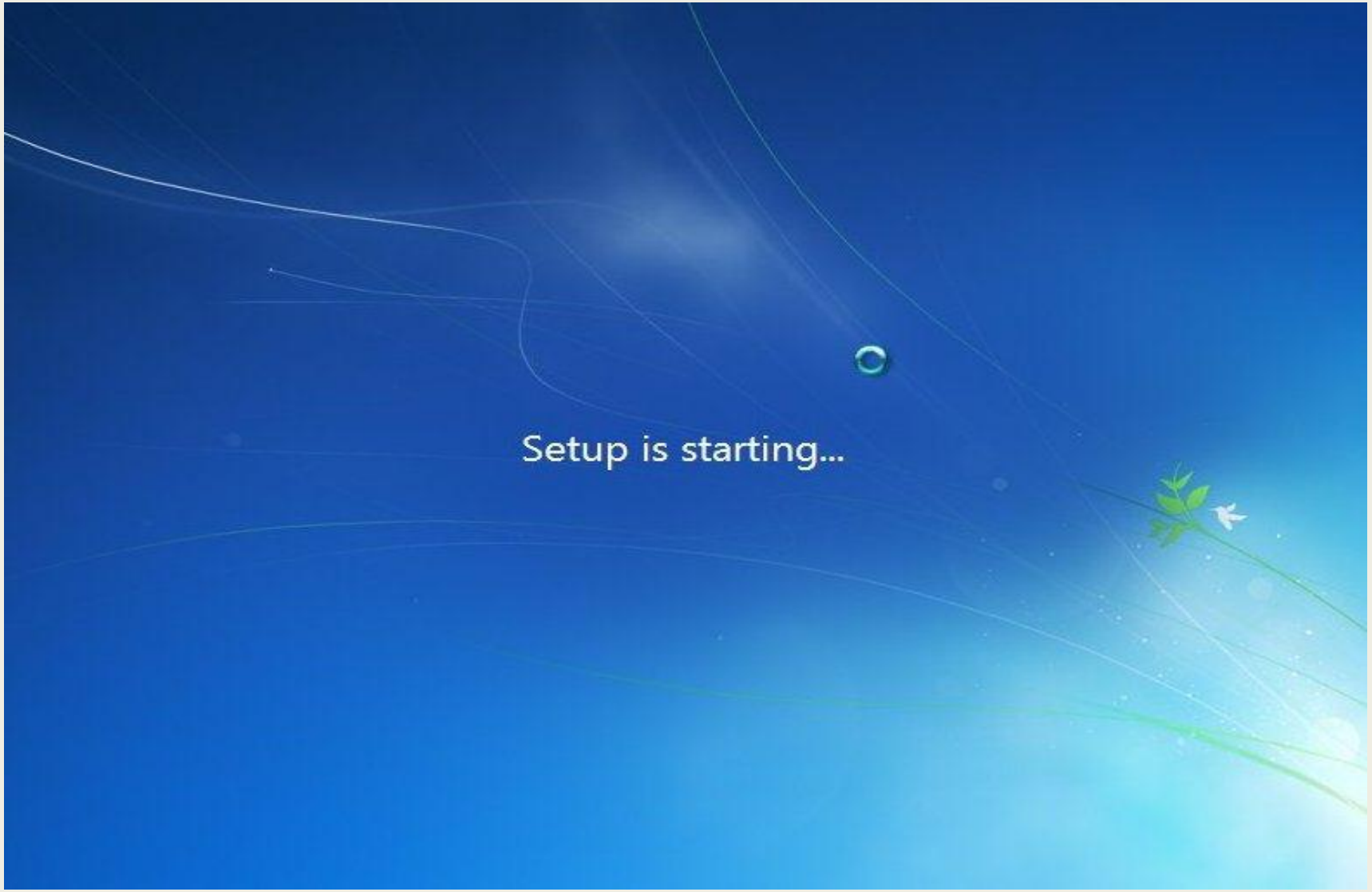


Select your language, time & currency format, keyboard or input method and click Next.

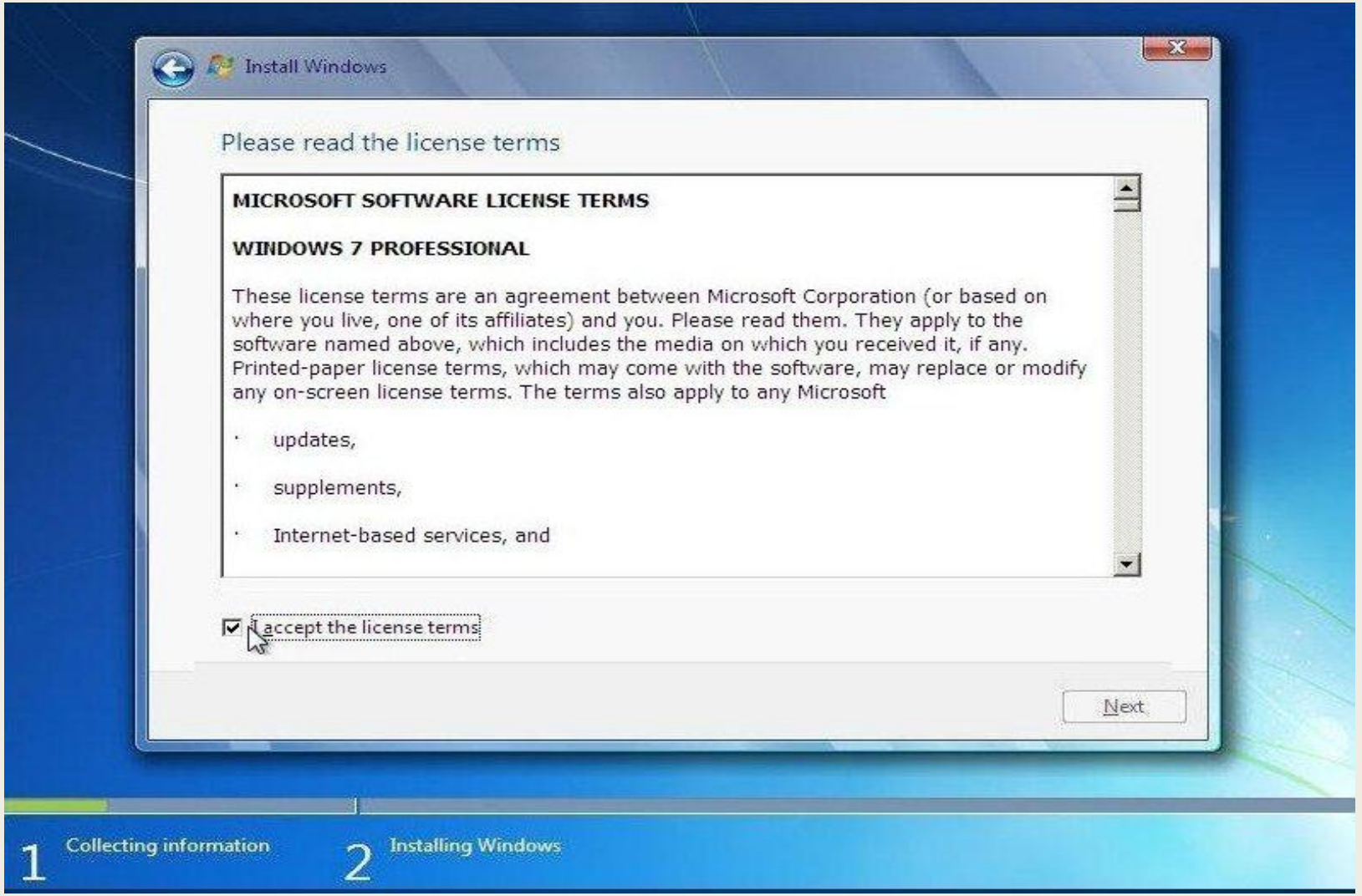


Click Install now.

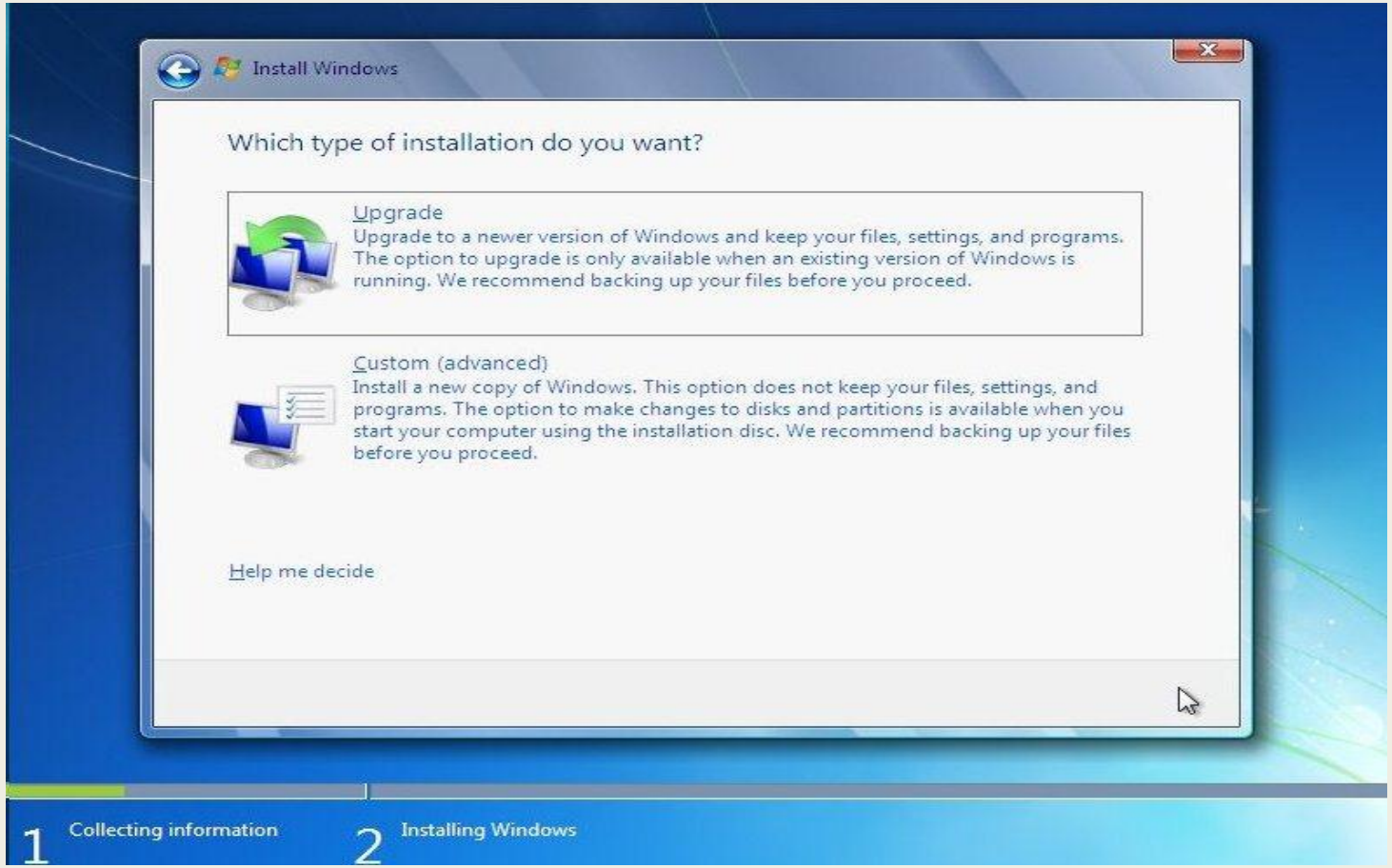



Setup is starting...

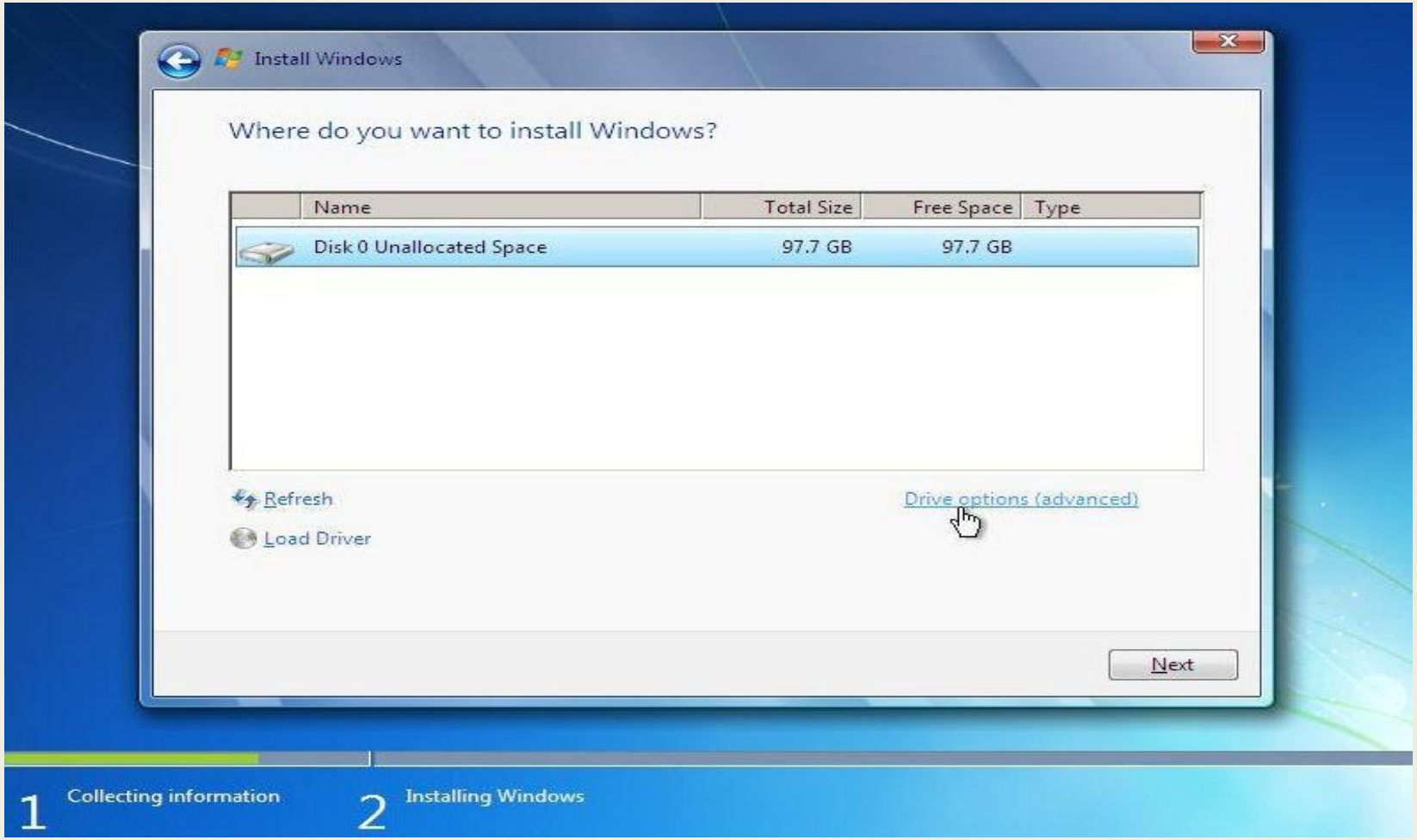
Check I accept the license terms and click Next.

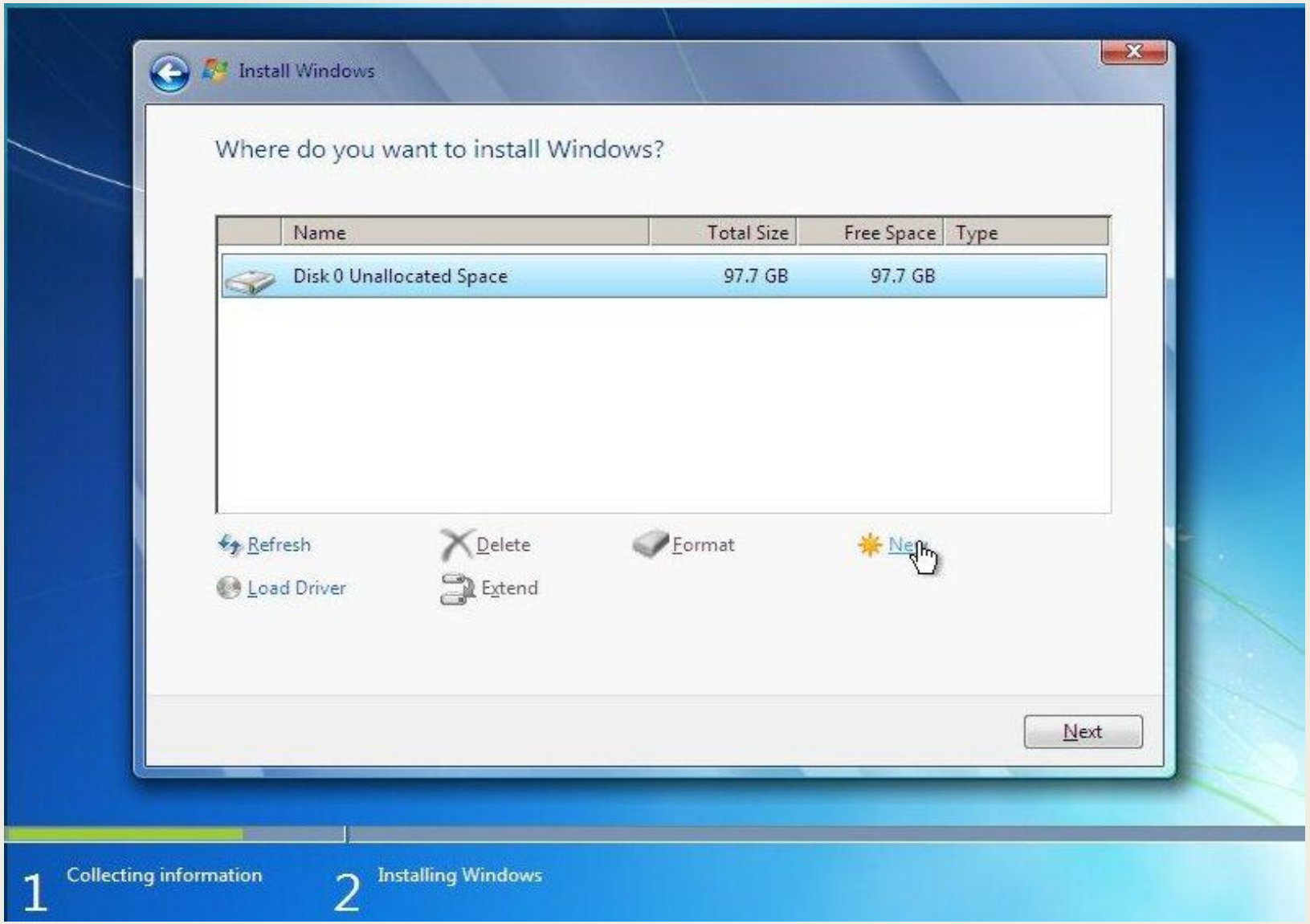


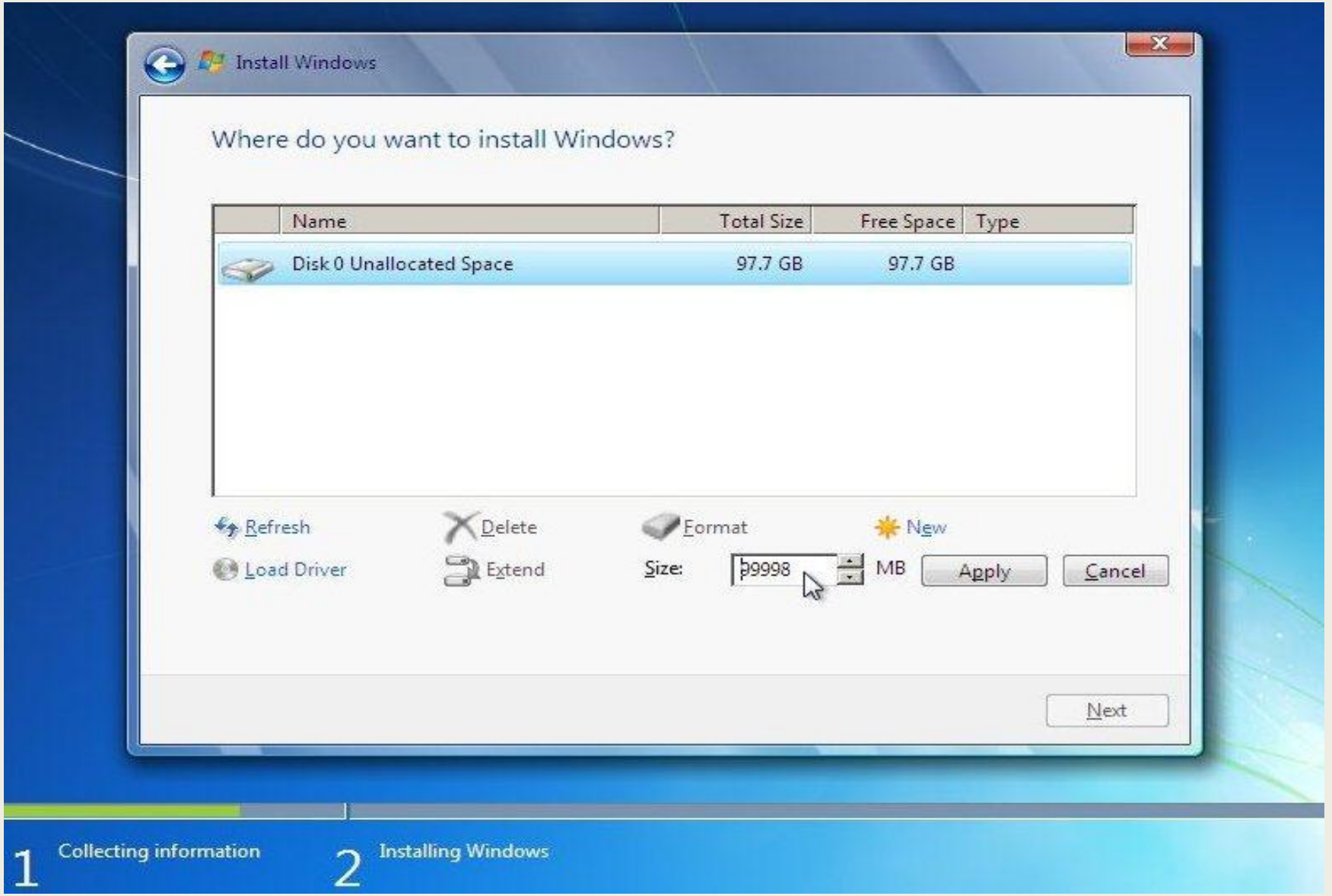
Click **Upgrade** if you already have a previous Windows version or **Custom (advanced)** if you don't have a previous Windows version or want to **install a fresh copy of Windows 7.**



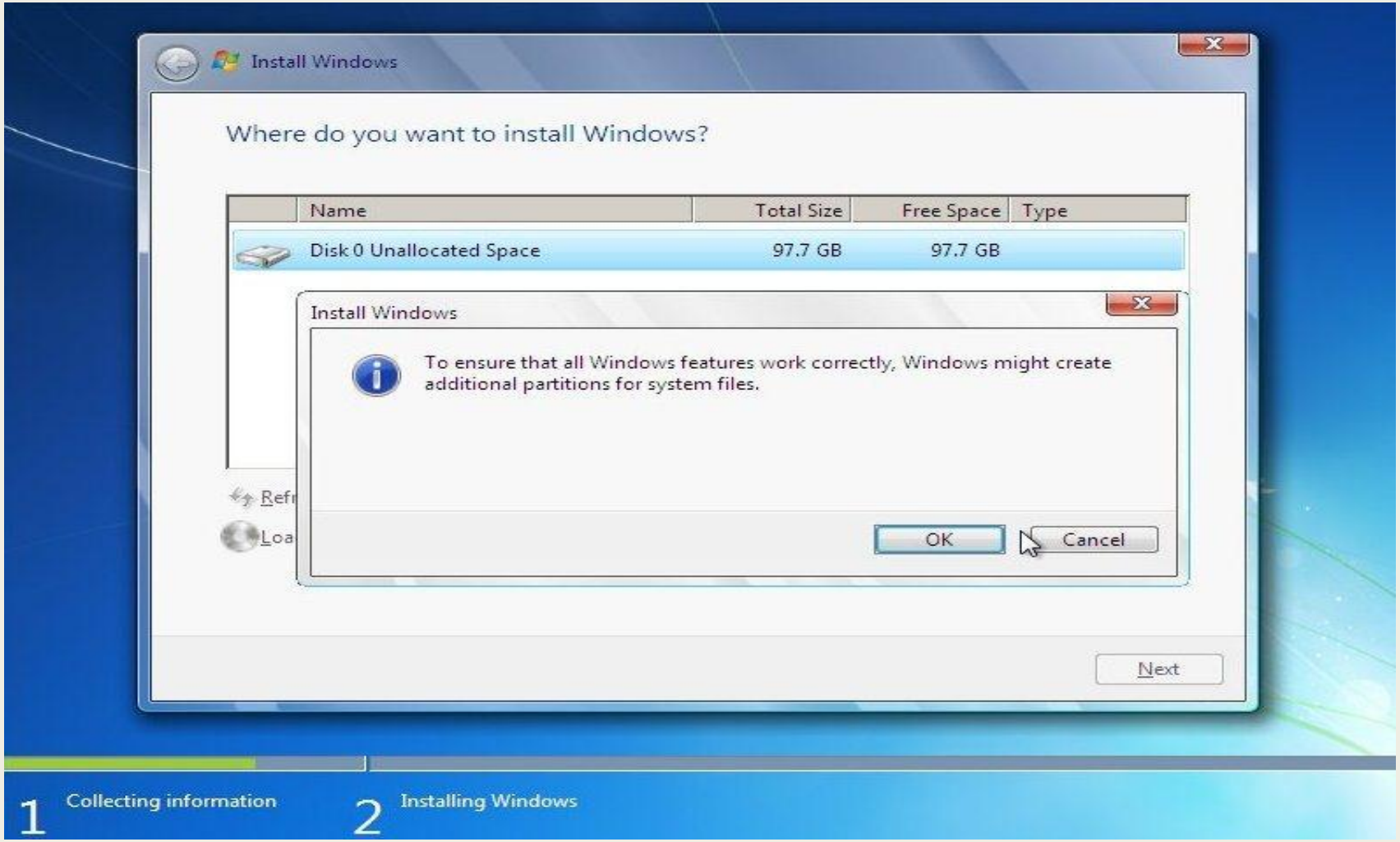
Select the Drive where you want to Install Windows 7 and click Next. If you want to make any Partitions, click Drive options (advanced), make the partitions and then click Next.

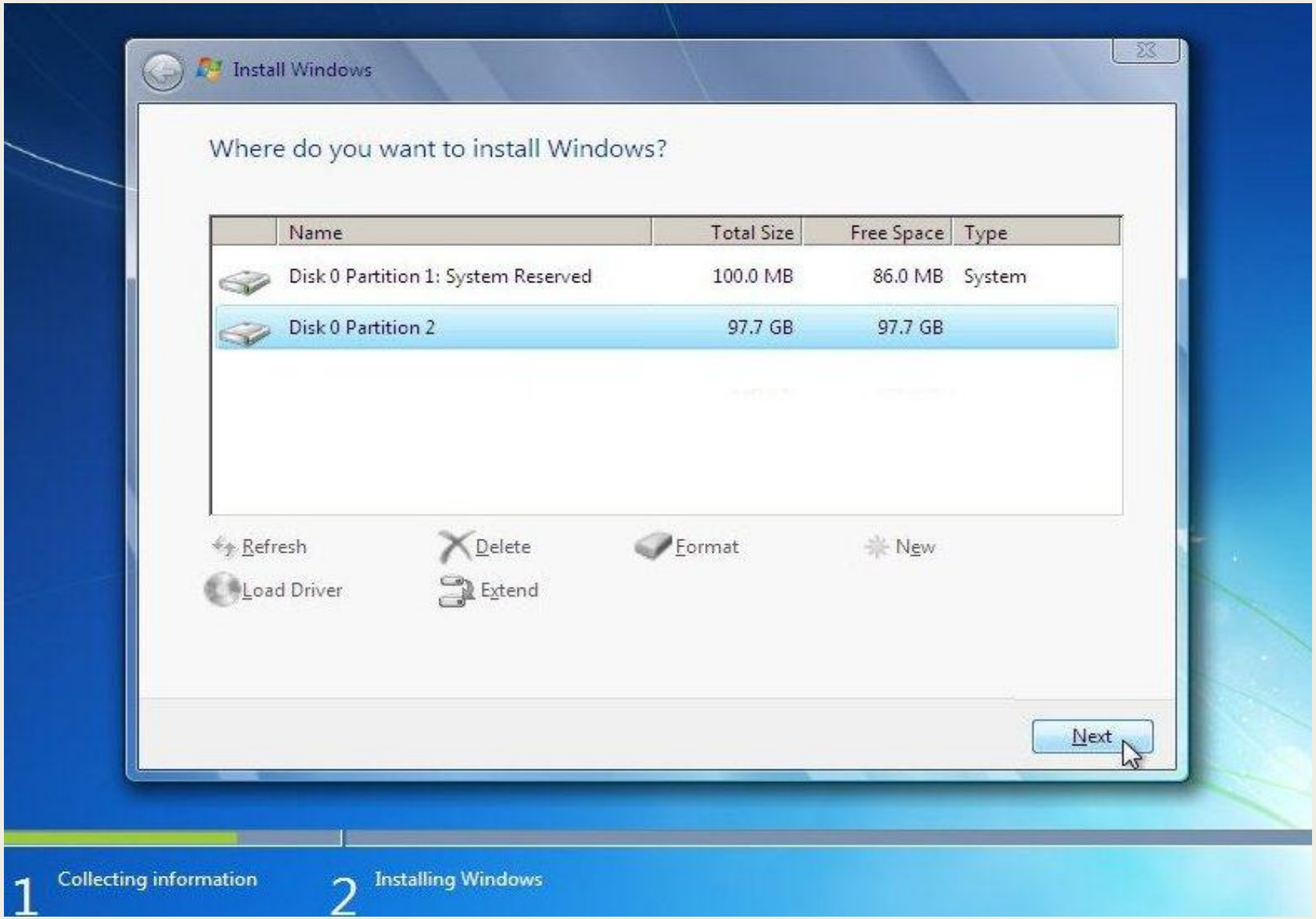



Install Windows
Where do you want to install Windows?
Name
Total Size
Free Space
Type
Disk 0 Unallocated Space
97.7 GB
97.7 GB
Refresh
Delete
Format
New
Load Driver
Extend
Next
1 Collecting information
2 Installing Windows

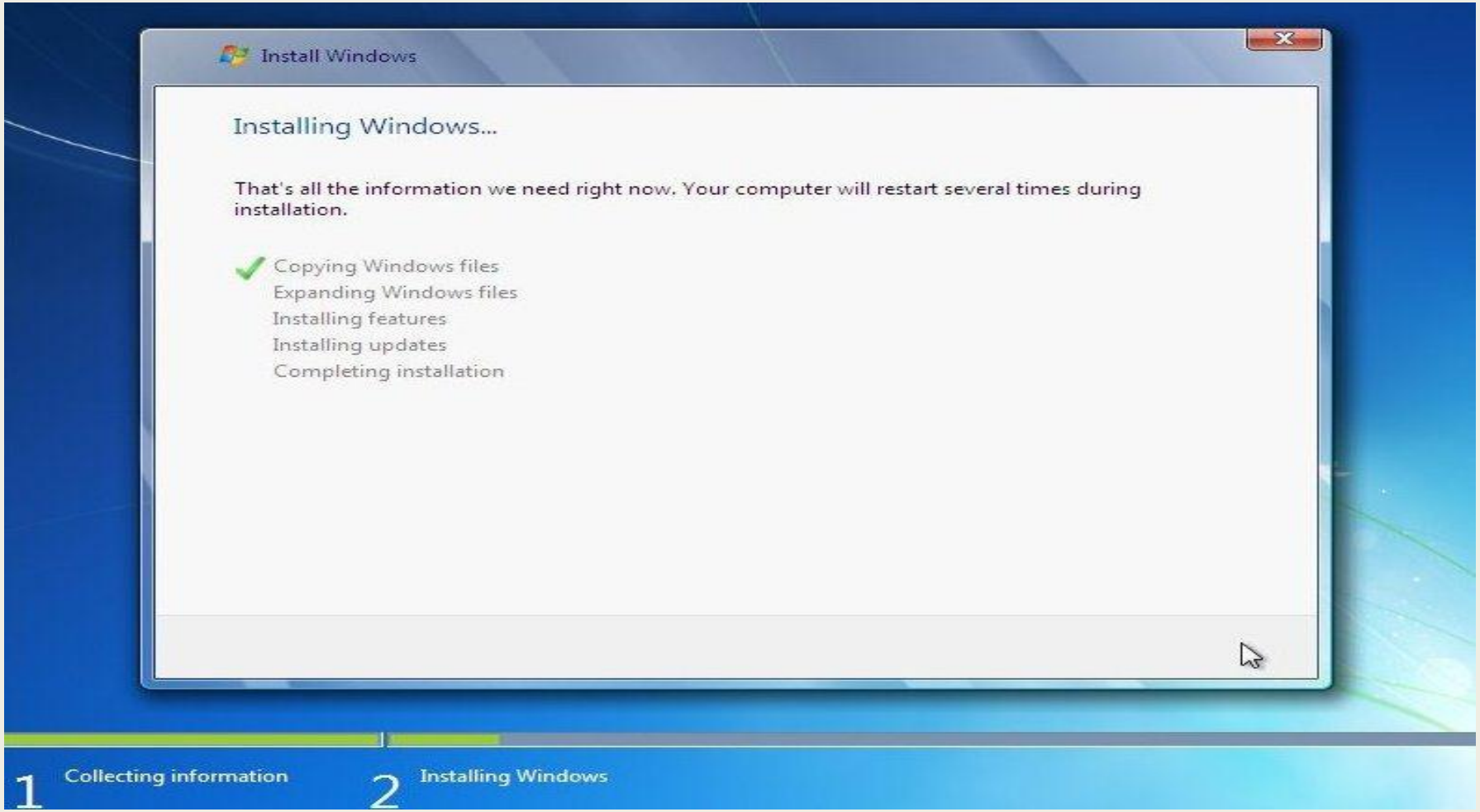

Install Windows
Where do you want to install Windows?
Name
Total Size
Free Space
Type
Disk 0 Unallocated Space
97.7 GB
97.7 GB
Refresh
Delete
Format
New
Load Driver
Extend
Size:
99998
MB
Apply
Cancel
Next
1 Collecting information
2 Installing Windows

Drive Options Menu: Windows warns that it may need to create a few Partitions for the system, Click "OK" to continue.

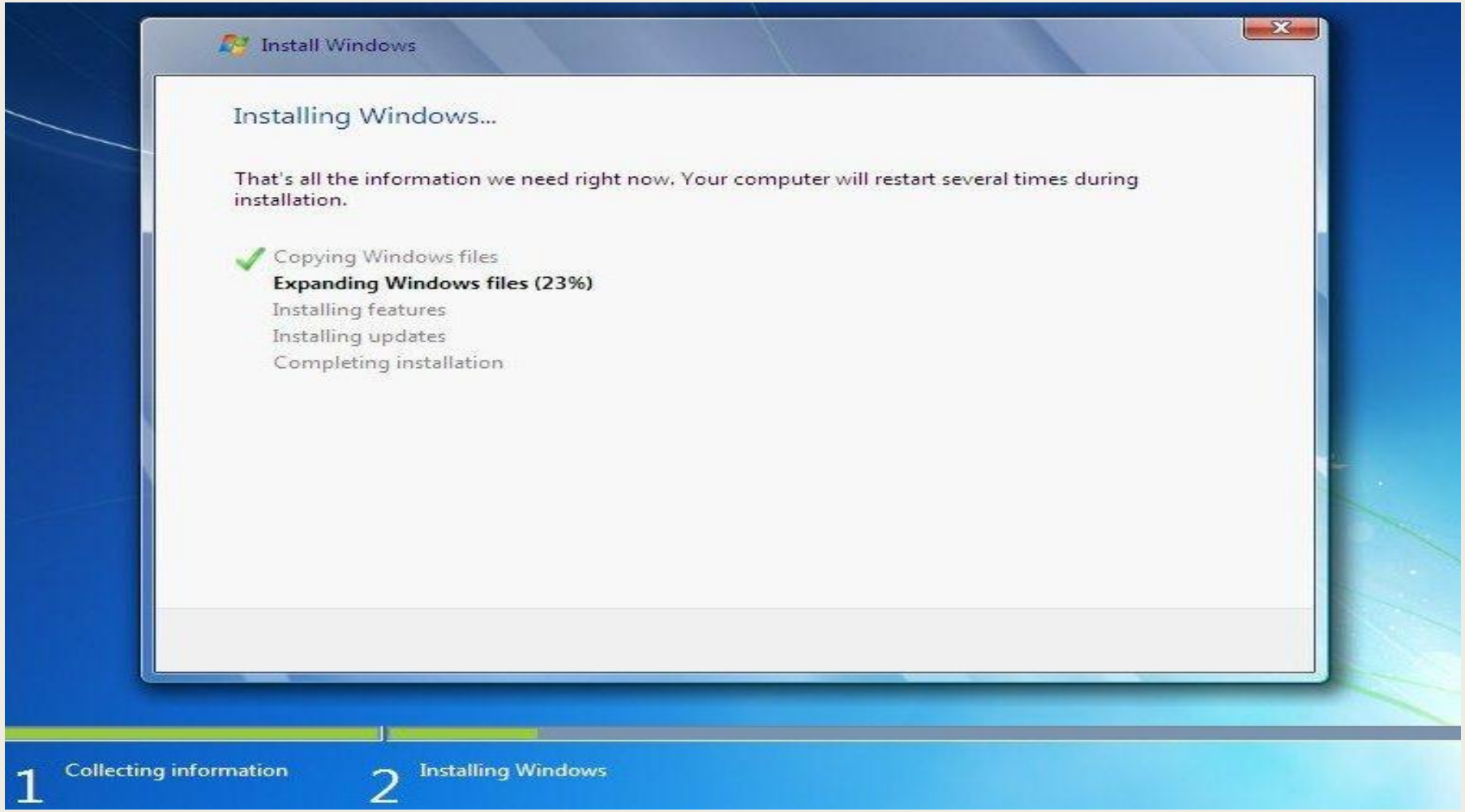


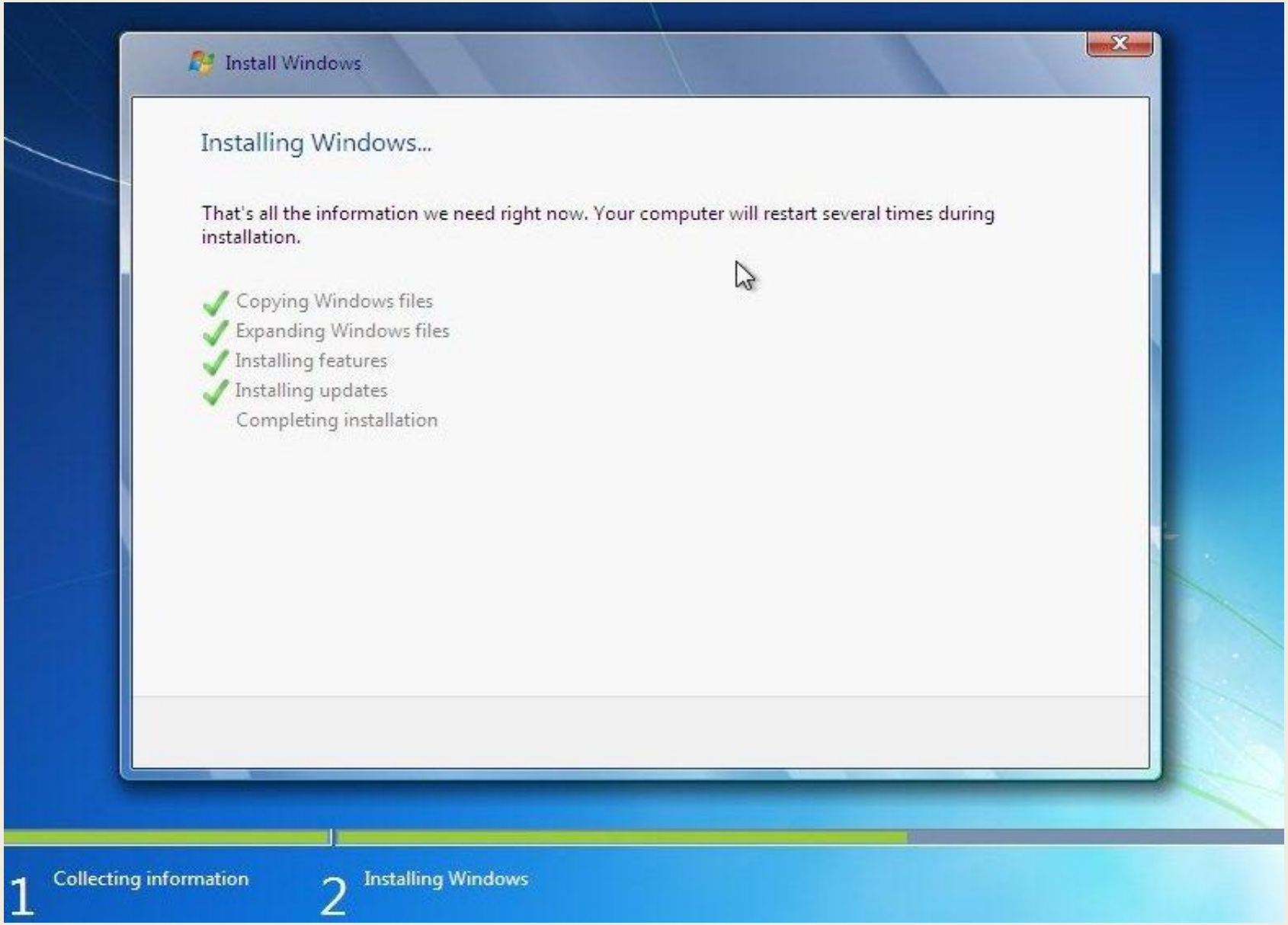

Install Windows
Where do you want to install Windows?
Name
Total Size
Free Space
Type
Disk 0 Partition 1: System Reserved
100.0 MB
86.0 MB
System
Disk 0 Partition 2
97.7 GB
97.7 GB
Refresh
Delete
Format
New
Load Driver
Extend
Next
1 Collecting information
2 Installing Windows

It will now start installing Windows 7. The first step, was already done when you booted the Windows 7 DVD so it will complete instantly.

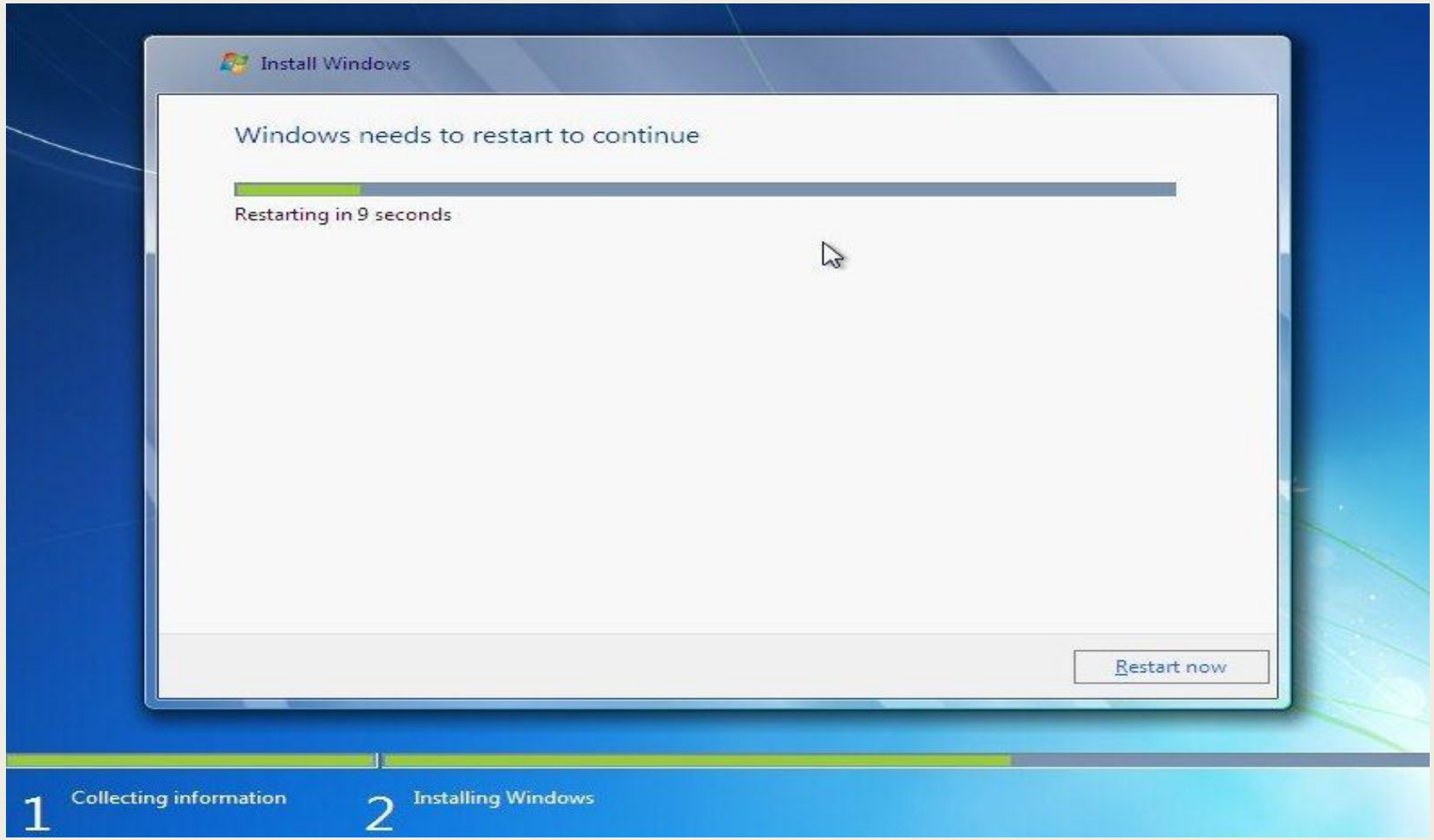


After completing the first step, it will expand (decompress) the files that it had copied.

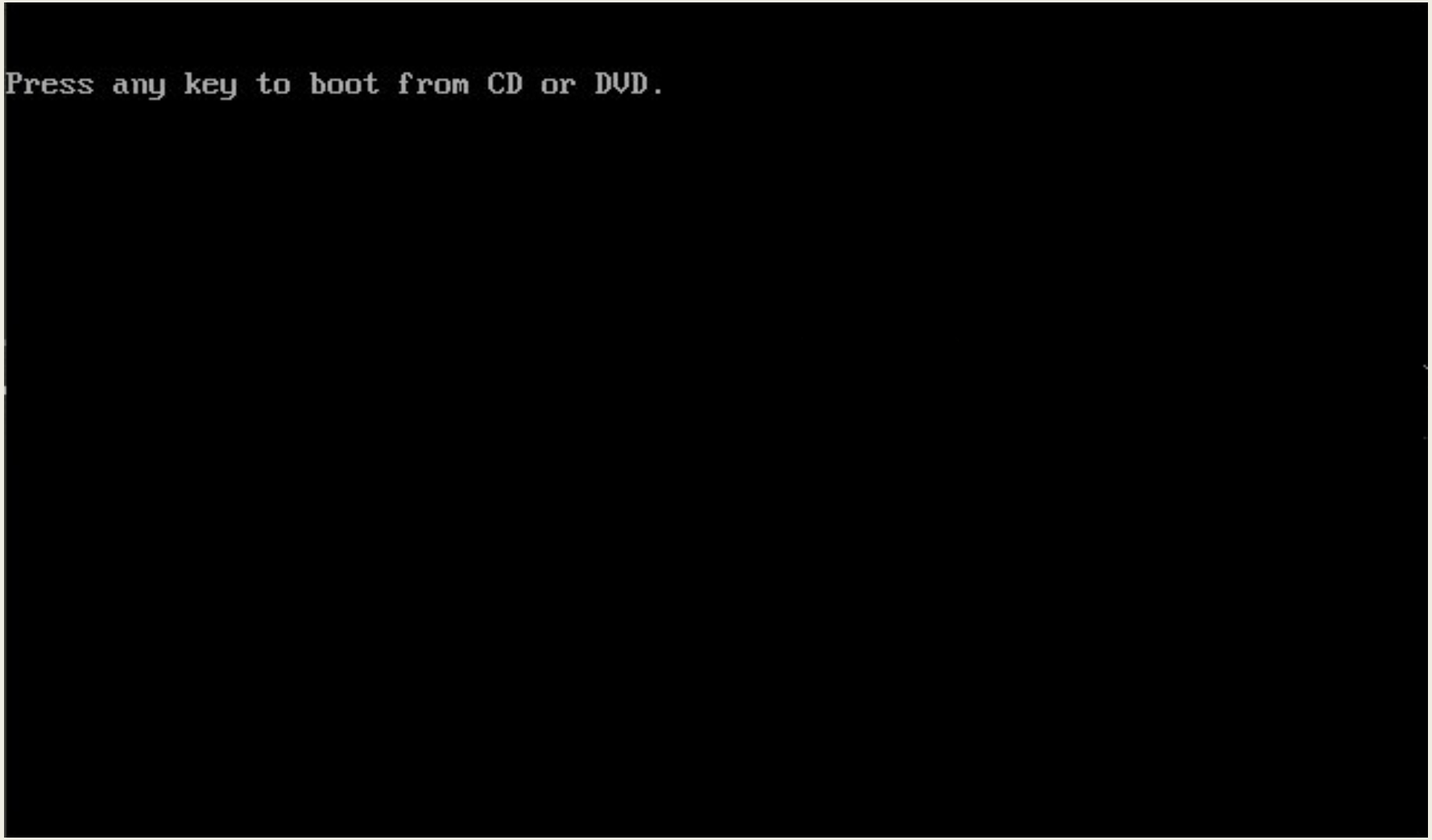


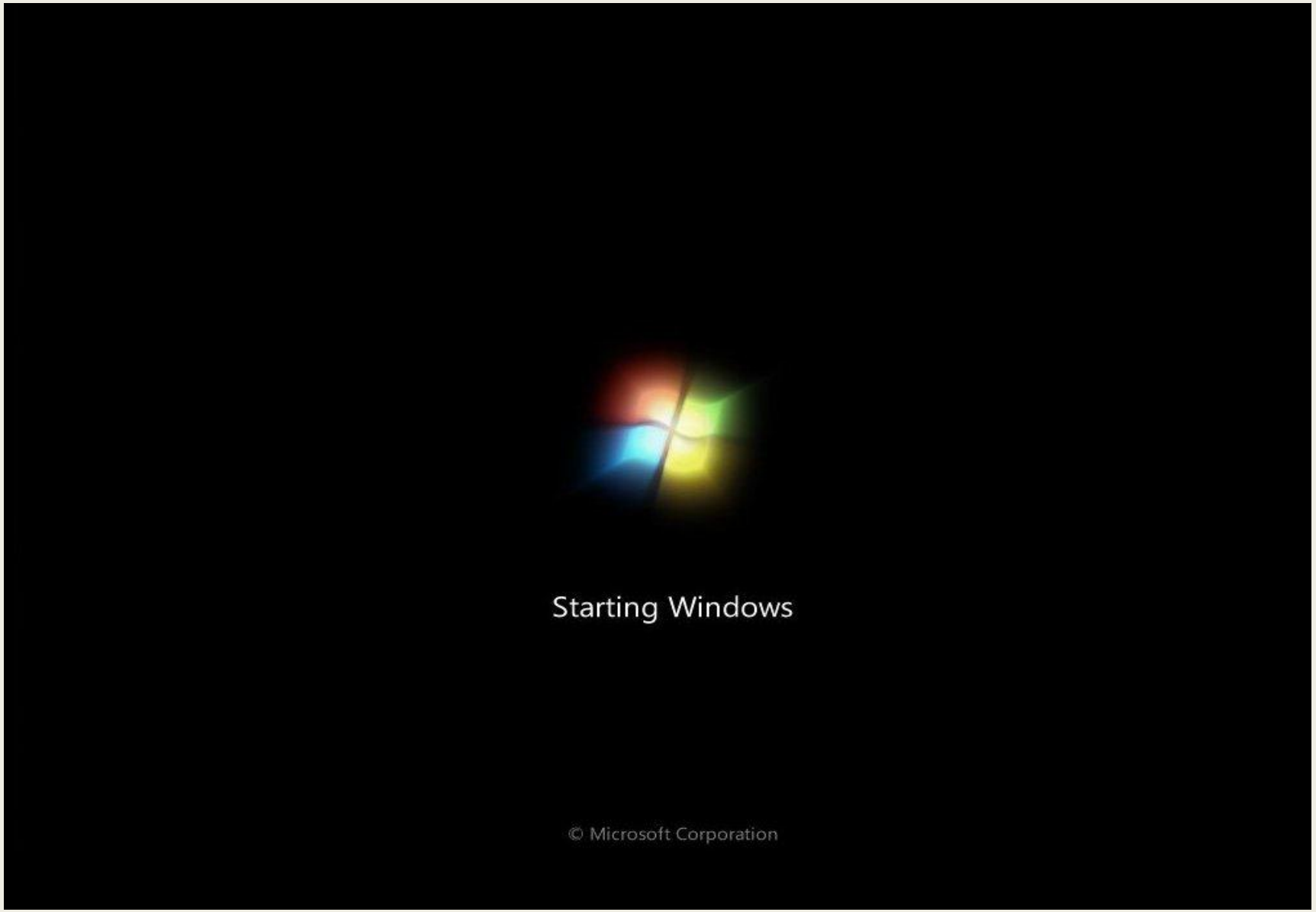

Install Windows
Installing Windows...
That's all the information we need right now. Your computer will restart several times during installation.
Copying Windows files
Expanding Windows files
Installing features
Installing updates
Completing installation
1 Collecting information
2 Installing Windows

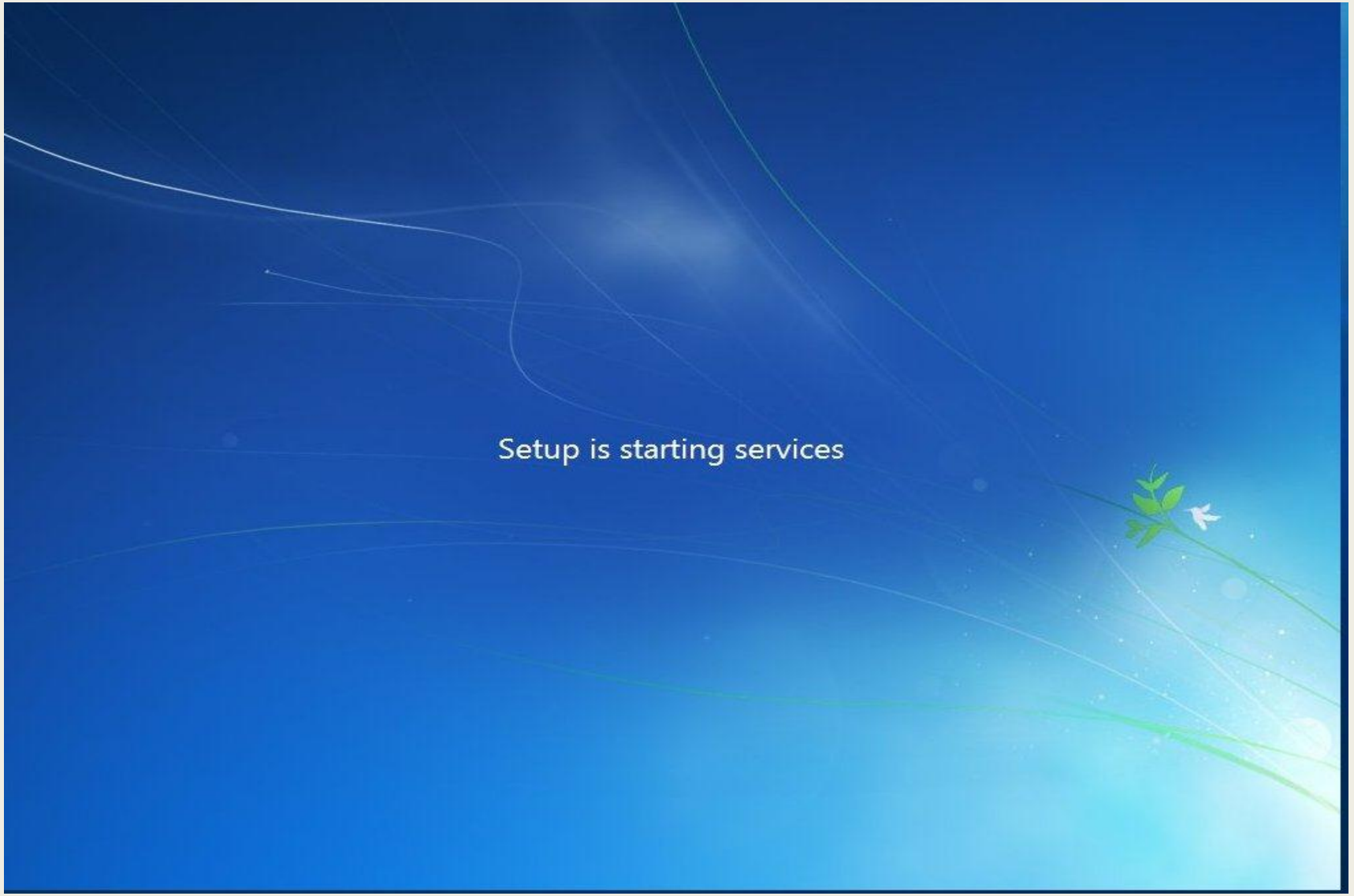
After that it will automatically restart after 15 seconds and continue the setup. You can also click Restart now to restart without any delays.



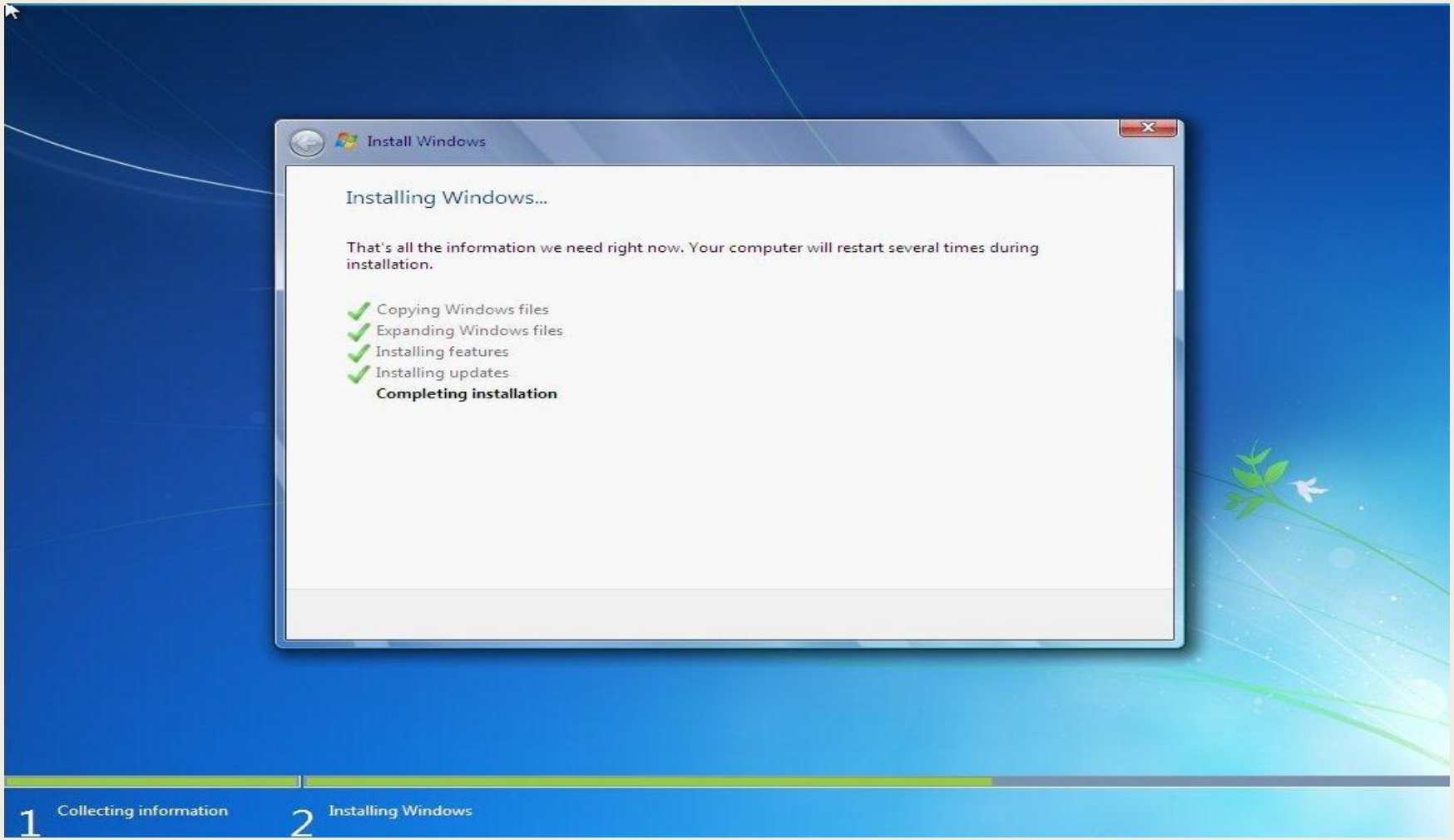
After installation, Please do not Press key this time

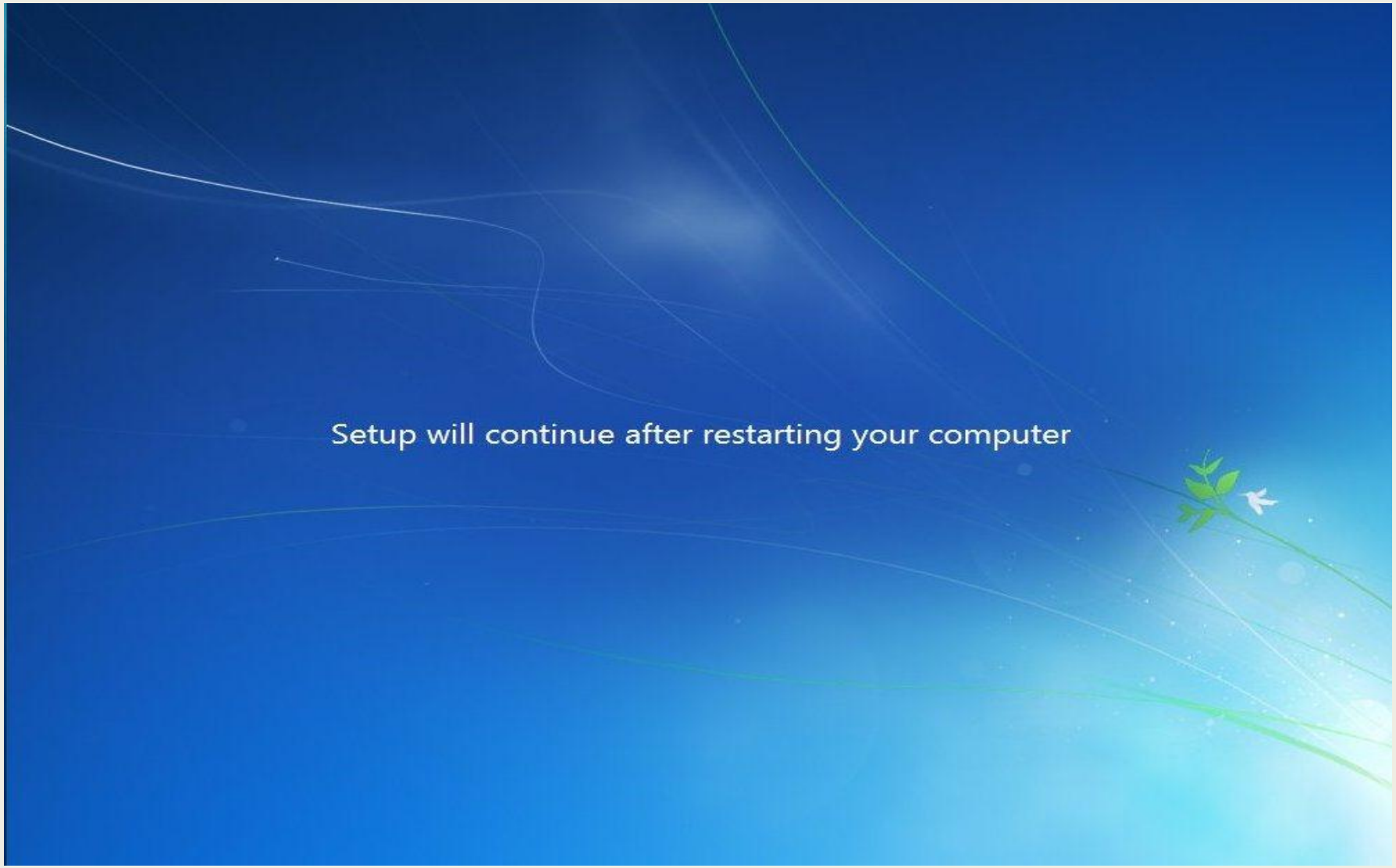



Starting Windows
© Microsoft Corporation

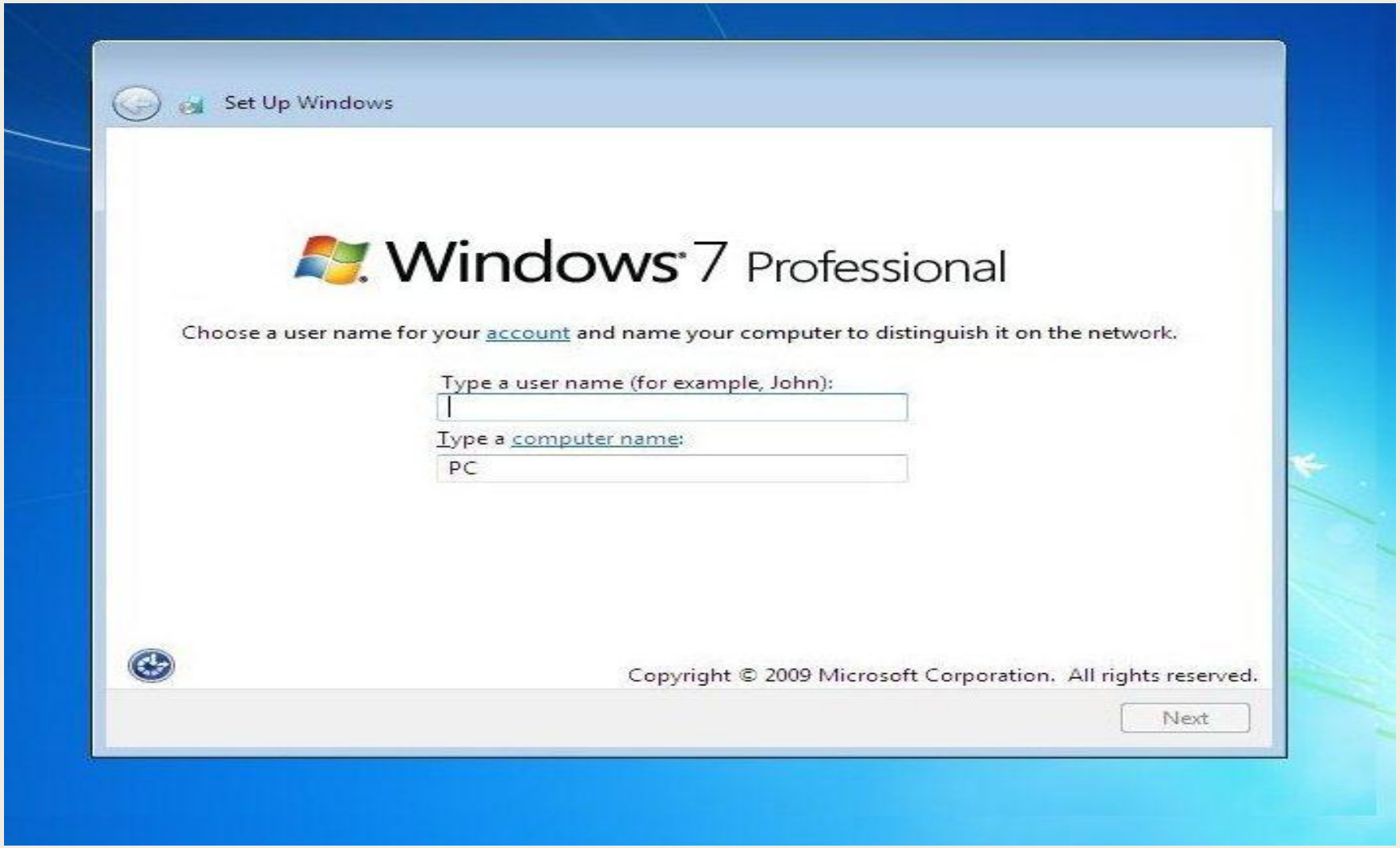

Setup is starting services

After Restarting for the first time, it will continue the setup. This is the last step so it will take the most time than the previous steps.




Setup will continue after restarting your computer

Type your desired User name in the text-box and click Next. It will automatically fill up the computer name.



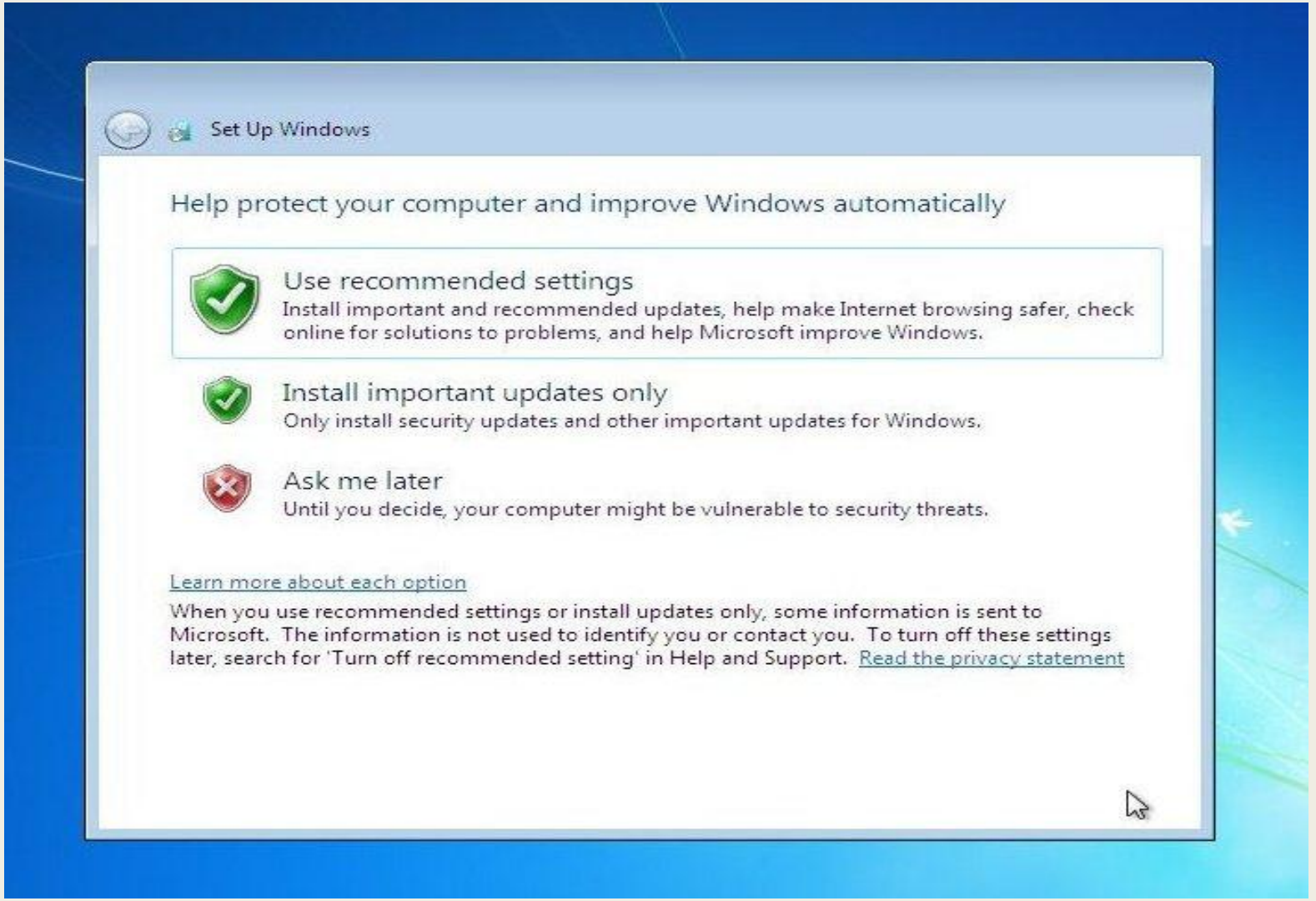
If you want to set a password, type it in the text-boxes and click Next.



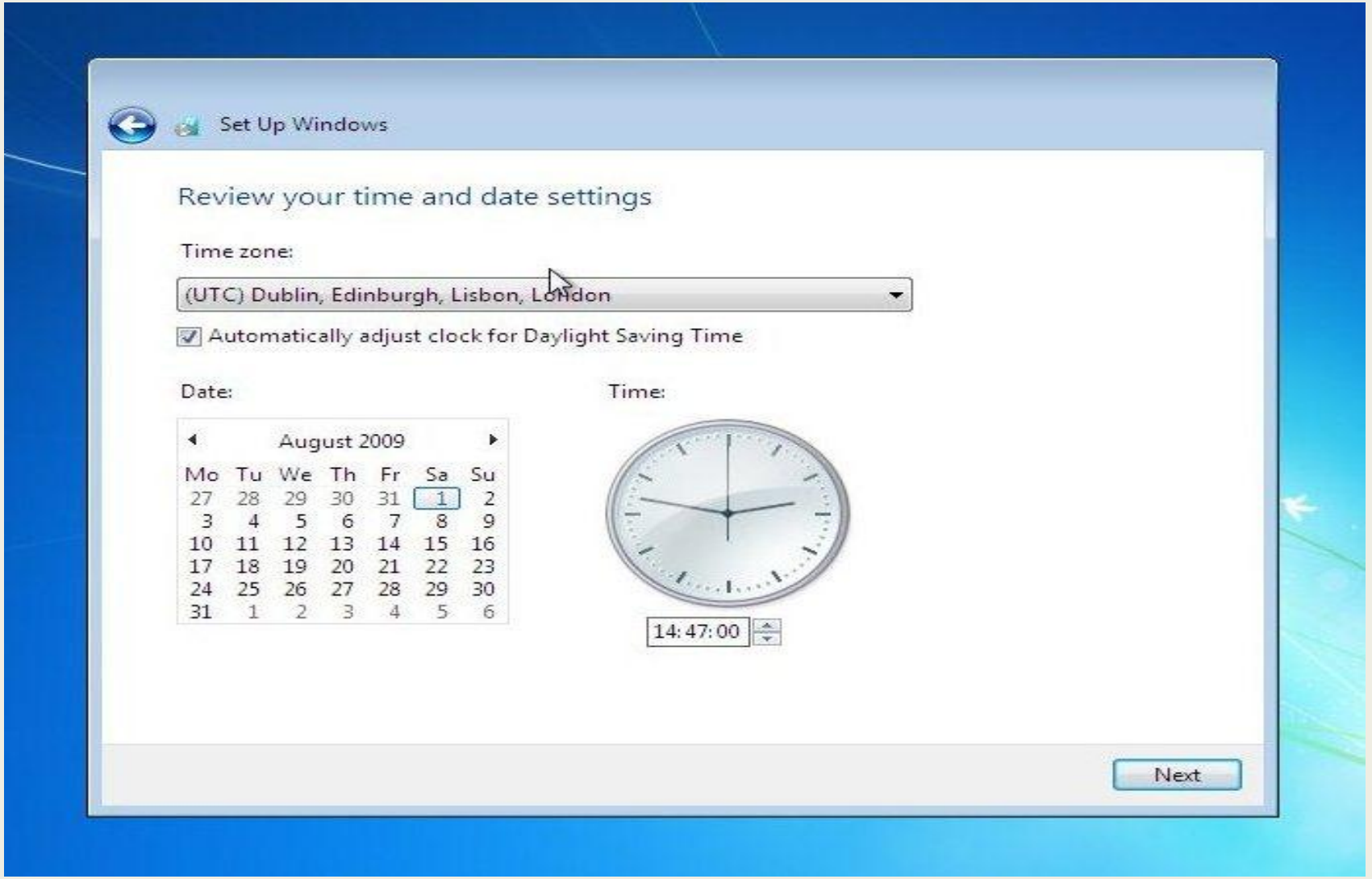
Type your product key in the text-box and click Next. You can also skip this step and simply click Next if you are using pre-activated windows.



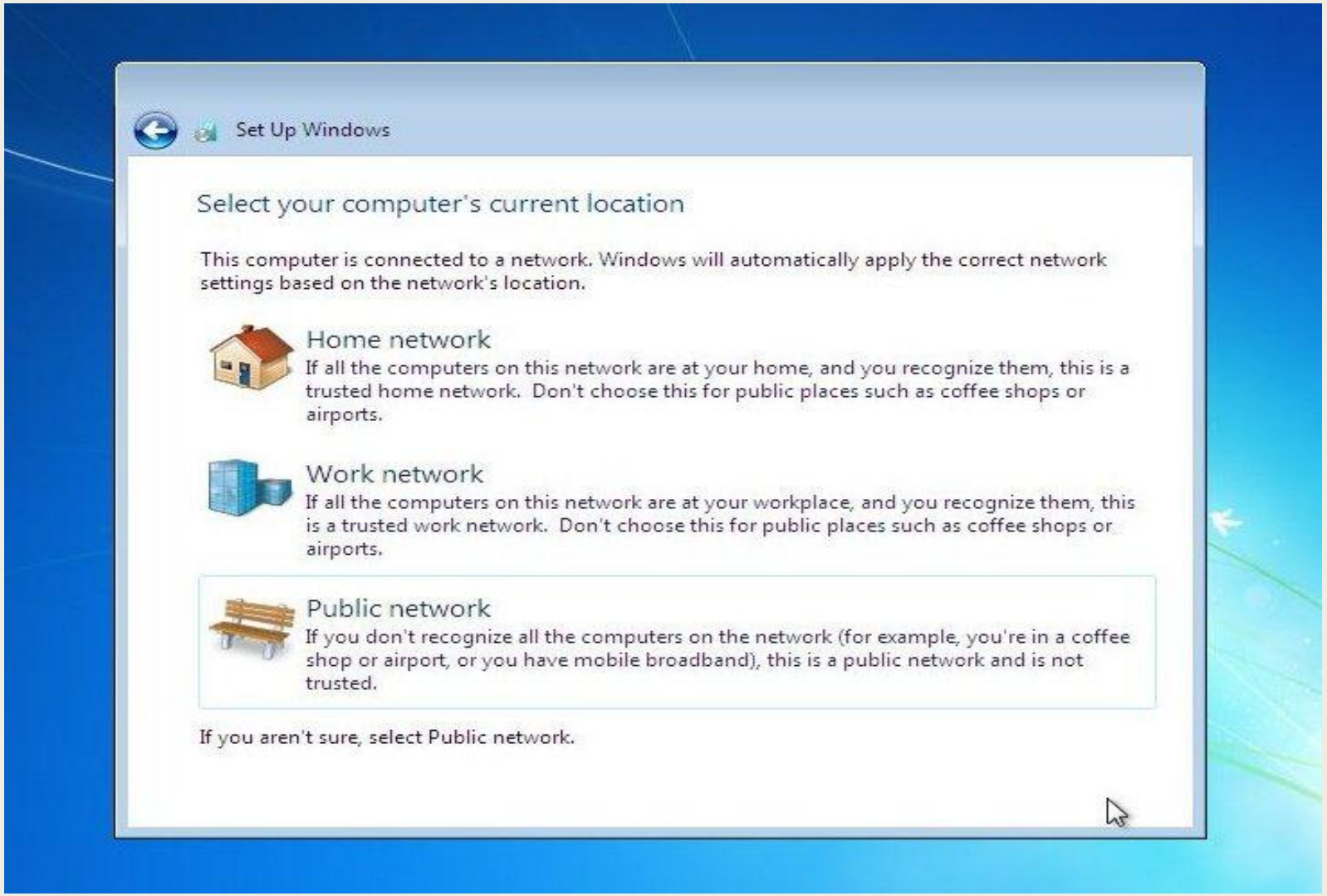
Select your desired option for Windows Updates.



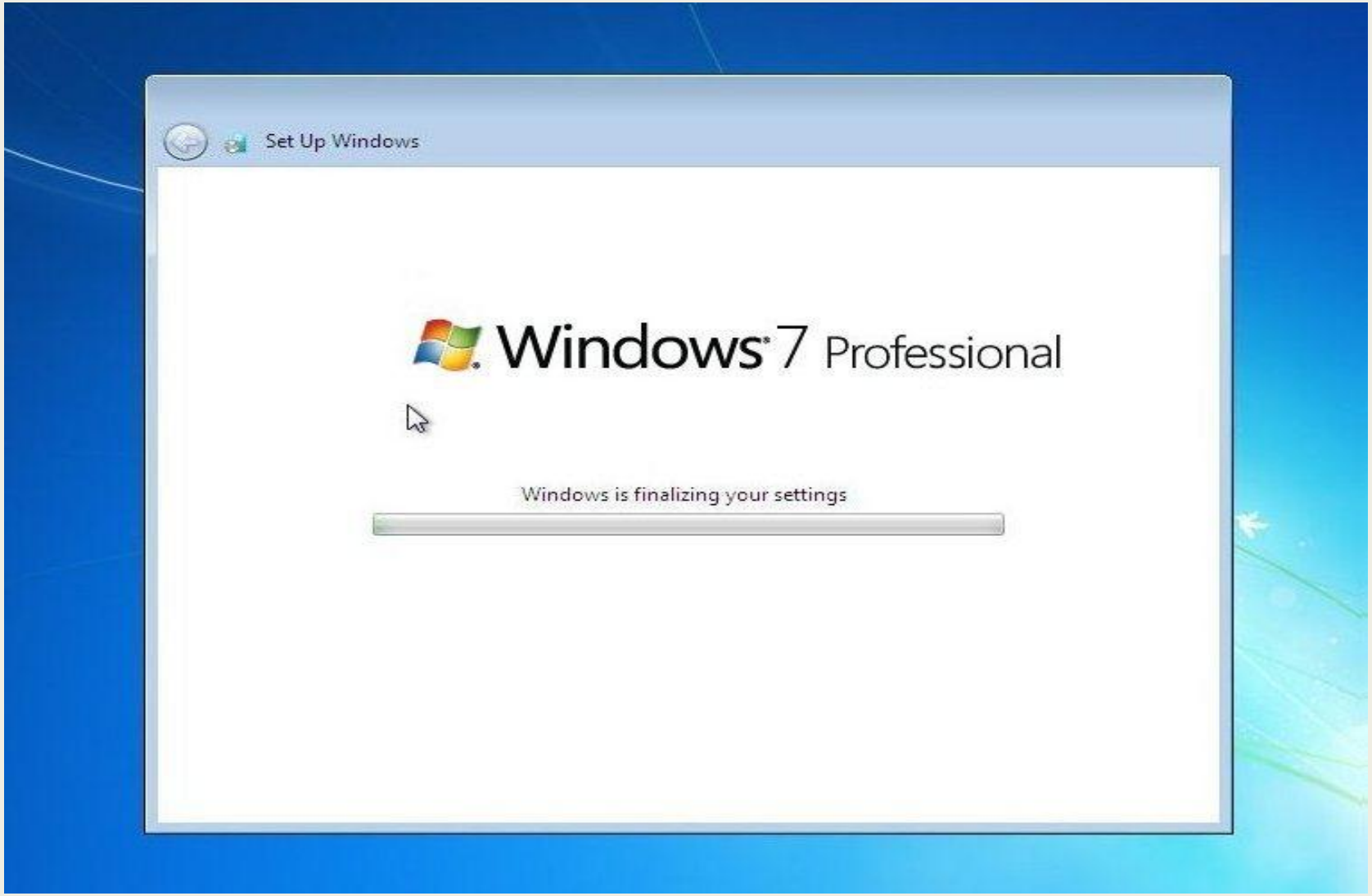
Select your time and click Next.



If you are connected to any network, it will ask you to set the network's location.



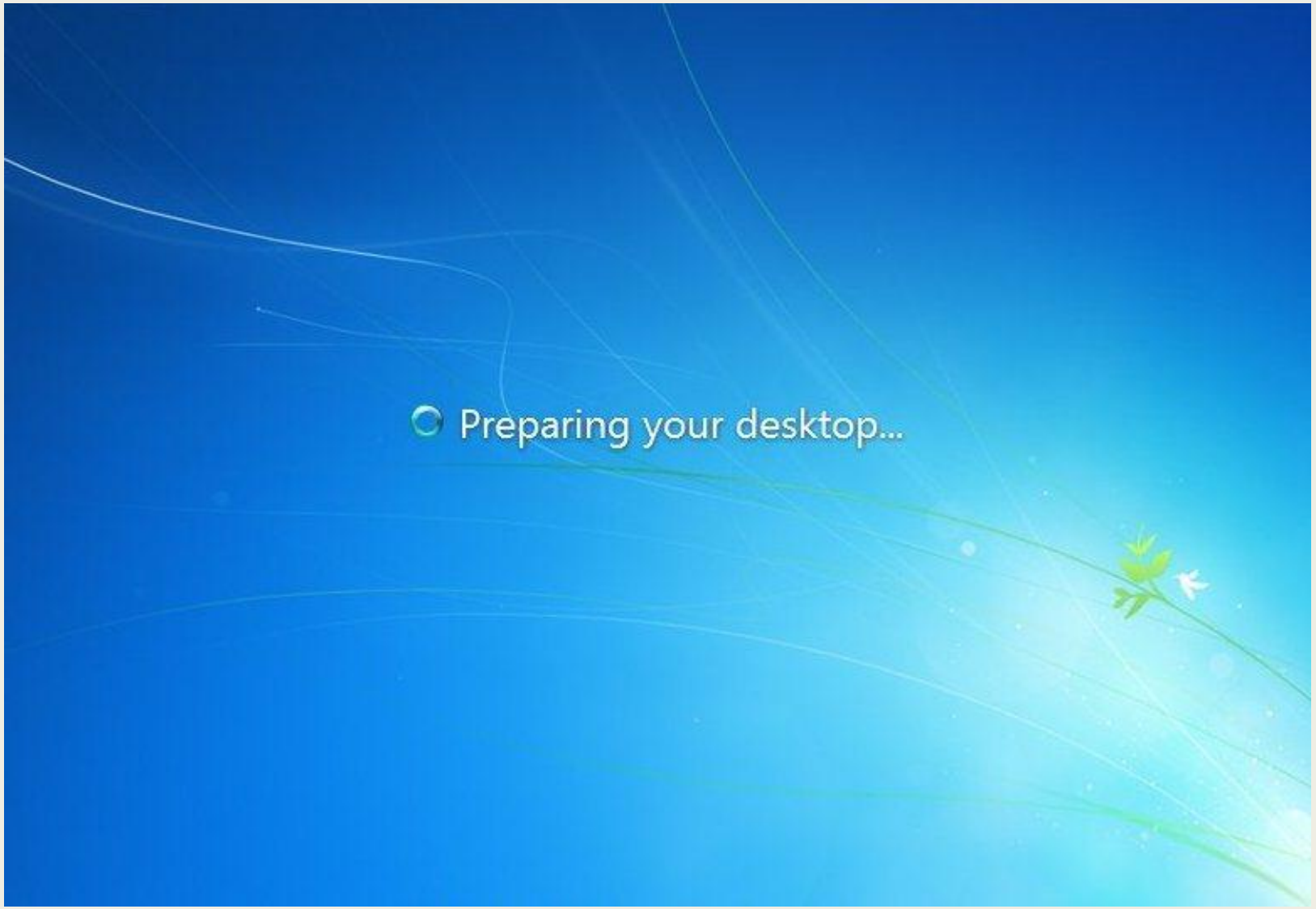
Window is finalizing your settings



Welcome Screen



Preparing your Desktop



And there you have a fresh copy of Windows 7 installed!

